Countercyclical Capital and Reserve Requirements as Macroprudential Stabilizers*

Diego Bohorquez† Alvaro Contreras‡

Bank of Italy Boston University

October 6, 2026

Abstract

This paper studies how reserve requirements and countercyclical capital requirements interact in stabilizing the business cycle. Using a small open-economy DSGE model with nominal and financial frictions, we characterize optimal simple rules for monetary and macroprudential policy under alternative central bank objectives. Reserve requirements mainly affect liquidity and funding conditions, while capital requirements act more directly on lending capacity and loan spreads, generating potentially different effects on credit, output, and inflation that matter for policy design. Both instruments can bring significant stabilization benefits when included in the policy mix, particularly once financial stability is part of the central bank's objective. When used separately to stabilize credit, capital requirements are more effective than reserve requirements. However, when both tools are used jointly but assigned distinct stabilization objectives, the optimal policy mix generally assigns reserve requirements to credit stabilization and capital requirements to output stabilization, although this assignment depends on the shocks hitting the economy. The results highlight gains from specialization across macroprudential instruments with distinct transmission channels.

*This paper previously circulated under the title "Countercyclical Capital and Reserve Requirements in a Small Open Economy." Diego Bohorquez thanks Priit Jeenas for his guidance on an earlier version of this project. We also thank Fernando Broner, Luigi Bocola, and Javier Bianchi for their insightful comments, as well as seminar participants at the Banca d'Italia & EIEF 4th Financial Intermediation Workshop, the CEMLA-LAJCB 2025 Conference, and CEBRA 2025. The views expressed are those of the authors and do not necessarily reflect those of the Bank of Italy.

†Bohorquez: Bank of Italy. Email: diego.bohorquez@bancaditalia.it.

‡Contreras: Boston University. Email: alvarocm@bu.edu.

Keywords: Macroprudential policy; Reserve requirements; Countercyclical capital buffers; Monetary policy; Financial stability; Small open-economy DSGE
JEL Codes: E44, E52, E58, E61, F41

1 Introduction

Recent policy discussions have increasingly emphasized the role of macroprudential policy not only in strengthening financial resilience, but also in stabilizing macro-financial fluctuations.[1] The macroprudential toolkit, however, includes instruments that operate through different margins of banks' balance sheets, notably liquidity and capital. How these instruments interact with one another, and with monetary policy, is therefore central to the design of macroprudential policy over the business cycle (Angeloni et al., 2024). This raises a natural question: how should liquidity- and solvency-oriented instruments be combined when they affect bank lending and economic activity through different transmission channels? We address this question by comparing reserve requirements with countercyclical capital requirements within a unified open-economy framework.

Both instruments are widely used worldwide, though with distinct geographic patterns. Reserve requirements have long been used to stabilize the credit cycle, although in recent years they have been employed primarily by emerging economies (Cordella et al., 2014; Cantú et al., 2024). Countercyclical capital requirements, by contrast, are a more recent instrument that gained traction in advanced economies following Basel III (see Figure 1). While each tool also serves broader purposes—liquidity management in the case of reserve requirements, and bank resilience in the case of capital requirements—we focus on their cyclical stabilization role.

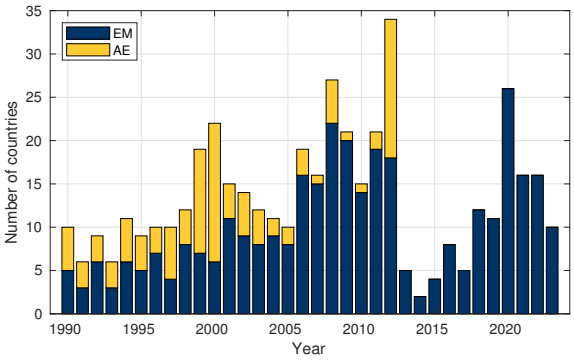
To address this question, we develop a DSGE model for a small open economy with nominal and financial frictions and a banking sector that collects deposits from households and extends loans to entrepreneurs. Banks are subject to both reserve and capital requirements, which operate through interest rate spreads and balance-sheet composition. This framework allows us to study the stabilization properties of each instrument and trace their effects on credit conditions, output, and inflation. We then characterize optimal monetary and macroprudential policy rules under alternative anchor variables and central bank objectives.

We contribute to the literature on macroprudential policy design by embedding two widely used instruments in a single general equilibrium framework. Doing so allows us to compare how each operates under a common benchmark, uncover potential complementarities that single-instrument analyses would miss, and examine how they interact with monetary policy in stabilizing the business cycle. This unified approach is particularly valuable given that evidence on the *joint* use of reserve and capital requirements is scarce, partly because few countries deploy both simultaneously. Moreover, the mech-

[1] See, for example, de Cos (2023) on the role of macroprudential policy in stabilizing macro-financial fluctuations and Bénassy-Quéré (2024) on the case for a macroprudential approach to bank liquidity.

anisms we identify extend beyond the specific instruments considered here, and could apply more broadly to other liquidity- and solvency-oriented macroprudential tools.

(a) Reserve Requirements tightening and/or loosening (# of countries)



(b) Countercyclical Capital Requirements tightening and/or loosening (# of countries)

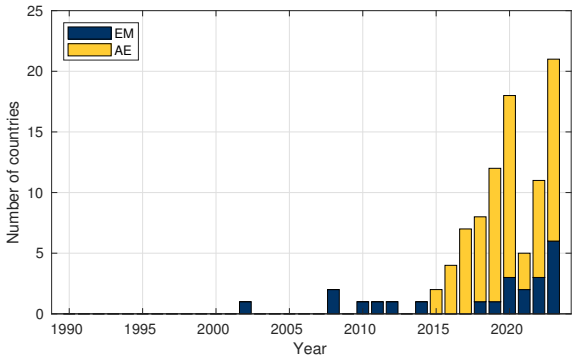


Figure 1: Reserve requirement and CCyB policy, by year

Note: EM = Emerging Markets, AE = Advanced Economies. The classification of countries is based on IMF's World Economic Outlook 2018. Panel (a) considers reserve requirements (domestic or foreign currency) for macroprudential purposes (although distinguishing them from those used for monetary policy purposes is often not clear-cut). Panel (b) considers a requirement for banks to maintain a countercyclical capital buffer (CCyB). Implementations at 0% are not considered as a tightening. ***Source:*** IMF's iMaPP database.

Three main lessons emerge from the analysis. First, the two instruments differ in their transmission, and therefore in their effects on the economy. In our calibration, tighter reserve requirements tend to raise inflation, while higher capital requirements reduce it. This asymmetry reflects the fact that the two tools operate through different margins of banks' balance sheets. Reserve requirements act mainly through liquidity and funding conditions, whereas capital requirements affect lending capacity, loan spreads, and investment more directly. Despite these differences, both instruments become useful additions to the policy mix once financial stability is part of the central bank's objective. In that case, reacting to credit fluctuations through the interest rate rule delivers limited gains, while using a dedicated macroprudential instrument is more effective.

Second, the preference for one instrument over the other, and the way in which the two tools can complement each other, are not always straightforward. When they are assigned to the same stabilization objective, capital requirements are generally the stronger instrument for mitigating credit volatility. However, this ranking does not determine how the two instruments should be assigned when the central bank prefers a strict separation of tasks, with one instrument responding to credit and the other to output. In this case, the model favors assigning reserve requirements to credit and capital requirements to output. The relevant comparison is therefore not only which instrument moves credit more strongly, but also how the two instruments can complement each other when assigned different roles.

Third, the preferred division of tasks is not universal and can change with the shock hitting the economy. Under a strict separation of tasks, the ranking reverses when we consider a government-spending or foreign-interest-rate shock in isolation, with reserve requirements responding to output and capital requirements to credit. The analysis also shows why macroprudential instruments can be particularly useful when macroeconomic and financial conditions call for different policy responses. Following an increase in government spending, for example, output and inflation rise while investment and credit contract, leading monetary policy to tighten while macroprudential policy eases.

Related Literature. Our paper builds on the macroprudential DSGE literature. First, it bridges two separate strands on macroprudential instruments. One focuses on countercyclical capital buffers and bank capital regulation (Mendicino et al., 2018; Malherbe, 2020; Darracq Pariès et al., 2022; Koch et al., 2025; Hecker et al., 2024), building on earlier work on the design and calibration of capital buffers (Darracq Pariès et al., 2011; Drehmann et al., 2011; Drehmann and Gambacorta, 2012; Beneš and Kumhof, 2015). Another studies reserve requirements and related liquidity tools, especially in emerging market and small open economy settings (Glocker and Towbin, 2012; Glocker and Towbin, 2015; Bustamante and Hamann, 2015; Agénor et al., 2018; Cordella et al., 2014; Cantú et al., 2024). These two branches have largely developed in isolation, with little work examining how the two classes of instruments interact when deployed simultaneously.[2]

A related literature studies how policy instruments should be assigned to stabilization objectives, typically through optimal simple rules following Schmitt-Grohé and Uribe (2007). Glocker and Towbin (2012) find that a separation of tasks—with monetary policy responding to inflation and output and reserve requirements responding to loans—can be advantageous. Frache et al. (2023) show that the source of shocks matters for the choice of activation variable and the relative performance of prudential tools. Whether this logic extends to a setting where two macroprudential instruments are available simultaneously, and whether the preferred assignment depends on the central bank's objectives, remains an open question.[3]

A growing body of work also examines the interaction between monetary and macroprudential policies. Since both transmit partly through the financial system, their interaction can generate complementarities as well as coordination problems (Cecchetti and Kohler, 2014; Laeven et al., 2022; Altavilla et al., 2020). In DSGE frameworks with financial frictions, De Paoli and Paustian (2017), Gelain and Ilbas (2017), and Van der Ghote (2021) study their joint design, while García Revelo and Levieuge (2022) show

[2]See Cerutti et al. (2017) for a review of the use of macroprudential instruments.

[3]Other applications of optimal simple rules to macroprudential policy include Angelini et al. (2014), Quint and Rabanal (2014), Clancy and Merola (2017), Rubio and Carrasco-Gallego (2014), Rubio and Carrasco-Gallego (2015), Ferreira and Nakane (2018), Lozej et al. (2023), and Hecker et al. (2024).

that some shocks can create conflicts between their mandates. Existing work, however, typically considers a single macroprudential instrument; how the monetary policy conflict changes when two instruments with asymmetric inflation transmission are available simultaneously has received little attention.[4]

The closest paper to ours is Carvalho and Castro (2017). They study a broader set of macroprudential instruments in a rich estimated DSGE model for Brazil, including reserve requirements and a countercyclical capital buffer. Our paper differs in focus: we abstract from some of that institutional detail in order to isolate the interaction between the two instruments that operate through different regulatory margins. In particular, while their optimal-policy exercise treats reserve requirements mainly as regulatory processes or shocks, we include them directly in the optimized macroprudential policy mix. This places reserve and countercyclical capital requirements on equal footing and allows us to study how each interacts with monetary policy.[5]

The rest of the paper is organized as follows. Section 2 describes the model and calibration. Section 3 studies how discretionary changes in reserve and capital requirements affect the banking sector and the economy. Section 4 discusses optimal policy rules. Section 5 examines the interaction between monetary and macroprudential policy when macroeconomic and financial conditions call for different policy responses. Section 6 studies how the preferred macroprudential assignments vary across shocks. Section 7 concludes.

2 The Model

We start from a standard small open-economy DSGE framework with investment and nominal rigidities. To this baseline, we add a banking sector with balance-sheet constraints and capital accumulation following Gerali et al. (2010), and we introduce reserve requirements in the spirit of Glocker and Towbin (2012). This setup, which includes several financial frictions, allows us to study how different macroprudential tools operate through bank balance sheets and interest rate spreads, and how they affect the transmission of shocks to the broader economy.[6]

Households consume domestic and foreign goods, supply labor, and save through bank deposits and internationally traded bonds. Firms produce differentiated interme-

[4] For related studies of policy interactions and multiple macroprudential instruments, see Agur (2019).

[5] A further difference is that Carvalho and Castro (2017) restrict countercyclical capital requirements to respond to the credit-to-GDP ratio, whereas we also consider rules that respond to the output gap, which yields a larger set of policy mixes.

[6] The model incorporates six frictions in total: Calvo-type price stickiness, investment adjustment costs, an external finance premium linking borrowing costs to entrepreneurial leverage and net worth, convex costs in holding bank capital, a reserve requirement wedge affecting deposit rates and interbank funding, and a net foreign asset adjustment cost.

diate goods under Calvo-type price stickiness, while entrepreneurs purchase capital and finance part of their investment through bank loans. Financial frictions therefore affect real activity through both savings decisions and credit conditions, allowing us to compare the impact of reserve requirements and capital requirements within a unified tractable framework. The model is solved by log-linearization around the steady state.

2.1 Banks

Banks are the central transmission channel for macroprudential policy in the model. To keep the two regulatory instruments analytically distinct, we represent each bank as comprising two branches. The deposit branch raises funds from households and allocates them between reserves at the central bank and interbank lending. The lending branch uses interbank funding together with bank capital to extend loans to entrepreneurs. This decomposition is convenient because reserve requirements and capital requirements operate on different parts of the intermediation process.

Both branches operate in perfectly competitive markets. Since banks are symmetric, their aggregate behavior can be characterized by a representative bank composed of a deposit branch and a lending branch.

2.1.1 Deposit Branch

The deposit branches collect deposits from households and allocate them between reserve holdings and interbank lending. Let D_t denote deposits, D_t^{IB} interbank lending, and Res_t reserves held at the central bank. The balance sheet of these branches is

$$Res_t + D_t^{IB} = D_t, \tag{1}$$

where $Res_t = \varsigma_t D_t$ and $D_t^{IB} = (1 - \varsigma_t)D_t$, while ς_t denotes the effective reserve ratio.

Deposits are remunerated at the gross deposit rate i_t^D, interbank lending earns the gross interbank rate i_t^{IB}, and reserves earn the gross reserve rate i_t^R. Deposit branches take the regulatory reserve requirement ratio τ_t^R as given and choose the composition of their assets accordingly. Deviations of the effective reserve ratio from the regulatory target are costly. We capture this cost with

$$G_t^{\varsigma} = \psi_1(\varsigma_t - \tau_t^R) + \frac{\psi_2}{2}(\varsigma_t - \tau_t^R)^2 \,, \tag{2}$$

where $\psi_1 < 0$ and $\psi_2 > 0$. The linear term allows banks to hold reserves slightly above the minimum in steady state, while the quadratic term disciplines fluctuations around the target. This specification captures, in reduced form, the idea that reserve management

becomes increasingly costly as banks move further away from the regulatory benchmark.

The deposit branches choose ς_t and D_t to maximize profits

$$\max_{\{\varsigma_t, D_t\}} \Pi_t^D = \left[(1-\varsigma_t) i_t^{IB} + \varsigma_t i_t^R - i_t^D - G_t^\varsigma\right] D_t. \tag{3}$$

The first-order conditions are

$$-(i_t^{IB} - i_t^R) - \psi_1 = \psi_2(\varsigma_t - \tau_t^R), \tag{4}$$

$$i_t^D = (1-\varsigma_t) i_t^{IB} + \varsigma_t i_t^R - G_t^\varsigma. \tag{5}$$

Equation (4) shows that the actual reserve ratio is lower when the opportunity cost of holding reserves $i_t^{IB} - i_t^R$ rises, and higher when the regulatory requirement tightens. The deposit rate is a weighted average of the returns on interbank lending and reserves, net of reserve-management costs. When reserve remuneration lies below the interbank rate, tighter reserve requirements lower the return banks can pass through to depositors. In that sense, reserve requirements act as a wedge on the funding side of the banking system.[7]

2.1.2 Lending Branch

The lending branches finance themselves through the interbank market from the deposit branches and with banking capital. They do not hold any deposits from households. Given this, they are not subject to reserve requirements, but they are subject to capital requirements.

Lending branches operate in perfectly competitive input and output markets. They supply loans to entrepreneurs at the lending rate i_t^L. This interaction is modeled through a financial contract as in Bernanke et al. (1999). The branch satisfies the balance-sheet identity

$$L_t = D_t^{IB} + K_t^b \,, \tag{6}$$

stating that it can finance loans L_t using either funds from deposit branches D_t^{IB}—at the cost of the interbank rate— or bank capital K_t^b. As in Gerali et al. (2010), the two sources of funding are perfect substitutes from the point of view of the balance sheet. Lending branches face costs related to the capital position of the bank. In particular, they pay a quadratic cost whenever the capital-to-assets ratio K_t^b/L_t moves away from a target value

[7] In order to solve the model, we will make some assumptions about how the central bank conducts monetary policy, which will lead to the reserve supply to adjust endogenously.

τ_t^K, set by the central bank. We model this cost as

$$\Upsilon_t = \frac{\kappa}{2}\left(\frac{K_t^b}{L_t} - \tau_t^K\right)^2 K_t^b, \tag{7}$$

where $\kappa > 0$. Bank capital is accumulated each period through retained earnings:

$$K_t^b = (1-\delta^b)K_{t-1}^b + \Pi_t^b\,, \tag{8}$$

where $\Pi_t^b = \Pi_t^L + \Pi_t^D$ denotes total banking-sector profits defined as the sum of lending and deposit branch profits, respectively.[8] δ^b measures resources spent in managing bank capital. Equation (8) assumes that all profits are retained in the banking sector, so any potential dividends that would otherwise accrue to households (or other agents in the model) are zero. Given this law of motion, bank capital is not a choice variable for the bank.

The problem the lending branch faces is to choose loans and interbank deposits to maximize profits,

$$\max_{\{L_t, D_t^{IB}\}} \Pi_t^L = i_t^E L_t - i_t^{IB} D_t^{IB} - \frac{\kappa}{2}\left(\frac{K_t^b}{L_t} - \tau_t^K\right)^2 K_t^b, \tag{9}$$

subject to the balance sheet in equation (6), where i_t^E denotes the effective lending rate on the aggregate loan portfolio.[9] The first-order conditions deliver the following expression:

$$i_t^E = i_t^{IB} - \kappa\left(\frac{K_t^b}{L_t} - \tau_t^K\right)\left(\frac{K_t^b}{L_t}\right)^2\,. \tag{10}$$

Equation (10) links the spread between the effective lending rate and the interbank rate to the capital position of banks. Absent the costs related to holding capital ($\kappa = 0$), banks would offer loans at the same interest rate they pay for the interbank deposits. However, with $\kappa > 0$, the spread is inversely related to the overall capital-to-assets ratio: when banks are less capitalized and leverage rises, lending margins widen. Intuitively, higher leverage tends to raise profits per unit of capital, but it also moves the bank further away from its target capital ratio and therefore increases the associated regulatory cost. The equilibrium spread reflects these opposing forces. This is the key channel through which countercyclical capital requirements affect credit conditions in the model,

[8] In equilibrium, however, $\Pi_t^D = 0$ as deposit branches operate in perfect competition and do not face any information frictions.

[9] The contractual lending rate i_t^L may be state-contingent. The effective lending rate i_t^E is the average return earned on the aggregate loan portfolio after accounting for borrower default. The spread condition for i_t^E comes from the ex-ante lending problem; but realized bank profits, and therefore bank-capital accumulation in (8), depend instead on the contractual lending rate i_t^L.

and it is precisely this margin that distinguishes their transmission from that of reserve requirements.

2.1.3 Equilibrium in the Banking Sector

In symmetric equilibrium, all deposit branches choose the same reserve ratio ς_t and set the same deposit rate i_t^D, while all lending branches face the same interbank rate and capitalization conditions. The consolidated banking-sector balance sheet is therefore

$$L_t = (1 - \varsigma_t)D_t + K_t^b. \tag{11}$$

Now, to understand how monetary policy works in the model, note that equation (4) can be written as follows:

$$\varsigma_t = \tau_t^R - \left(\frac{i_t^{IB} - i_t^R + \psi_1}{\psi_2} \right). \tag{12}$$

Denote by Δ_t the spread between the interbank rate and the rate paid on reserve balances: $\Delta_t \equiv i_t^{IB} - i_t^R$. Following Glocker and Towbin (2012), we assume that the central bank maintains Δ_t equal to a constant $\Delta \geq 0$, which from equation (4), pins down the difference between effective reserves and required reserves to a constant[10]

$$\varsigma_t - \tau_t^R = -\left(\frac{\Delta + \psi_1}{\psi_2} \right) \equiv \Omega. \tag{13}$$

Now, while the spread between the interbank rate and the rate on reserves is constant, the spread between the rate paid on deposits and the interbank rate is determined by the zero-profit condition for deposit-taking banks:

$$i_t^D = (1 - \varsigma_t) i_t^{IB} + \varsigma_t i_t^R - G^\varsigma = i_t^{IB} - \Delta \tau_t^R - \Omega\Delta - G^\varsigma \tag{14}$$

where G^ς is the cost of holding reserves from equation (2), that is now also constant given Δ and Ω. Equation (14) shows that changes in the reserve requirements will have a negative effect on deposit rates, scaled by the spread between the interbank rate and the rate paid on reserves.

Because households do not hold cash, aggregate nominal reserves $\varsigma_t P_t D_t$ correspond

[10] When $\Omega \geq 0$, banks hold reserves at least equal to the regulatory requirement. This restriction is useful for the economic interpretation of the steady state, but the quantitative results do not depend on the particular value of Ω, since the model is solved in deviations from the steady state.

to the monetary base. Real seigniorage revenue is therefore

$$T_t^S = \varsigma_t D_t - \frac{i_{t-1}^R}{\pi_t} \varsigma_{t-1} D_{t-1} \ ,$$

and is rebated lump-sum to households.

2.2 Entrepreneurs, Capital, and Credit

Entrepreneurs connect financial conditions to real activity. They acquire capital, rent it to intermediate-goods producers, and finance investment partly with internal net worth and partly with bank borrowing. As in standard financial accelerator environments, external finance becomes more costly as leverage rises, so changes in lending conditions feed directly into investment and, through capital accumulation, into future production.

2.2.1 Capital Producers

Capital goods producers purchase undepreciated capital from entrepreneurs and combine it with investment goods to produce the next period's capital stock. Investment is subject to quadratic adjustment costs,

$$\frac{\chi}{2}\left(\frac{I_t}{K_{t-1}} - \delta\right)^2 K_{t-1},$$

where δ is the depreciation rate and χ governs the sensitivity of the price of capital to investment fluctuations.

Let Q_t denote the market price of capital. The capital producer solves

$$\max_{I_t} \left[(Q_t - 1) I_t - \frac{\chi}{2}\left(\frac{I_t}{K_{t-1}} - \delta\right)^2 K_{t-1} \right]. \tag{15}$$

The first-order condition yields the capital supply curve $Q_t = 1 + \chi\left(\frac{I_t}{K_{t-1}} - \delta\right)$, and the aggregate capital stock evolves according to $K_t = (1-\delta)K_{t-1} + I_t$.

2.2.2 Entrepreneurs

Entrepreneurs purchase capital at the beginning of the period, rent it to intermediate-goods producers, and resell the undepreciated portion at the end of the period. Their expenditure on capital is financed through net worth N_t and bank loans, such that $Q_t K_t = N_t + L_t$.[11]

[11] We consider only the case in which the loan is denominated in domestic currency.

The interaction between entrepreneurs and the lending branch follows the financial-contract structure of Bernanke et al. (1999). Projects are subject to idiosyncratic shocks that are privately observed by entrepreneurs, while lenders can verify realized returns only at a cost. The optimal contract delivers a relationship between firms' leverage and the external finance premium:

$$Q_t K_t = f\left(\frac{E_t r^K_{t+1}}{i^E_t / E_t \pi_{t+1}}\right) N_t, \qquad f'(\cdot) > 0. \tag{16}$$

In the present framework, the financing cost for the lending branch is not simply the interbank rate, but is determined by bank capitalization through (10). This is the point at which the capital-regulation block and the financial accelerator interact. From equation (16), the external finance premium is

$$\frac{E_t r^K_{t+1}}{\left(i^{IB}_t - \kappa\left(\frac{K^b_t}{L_t} - \tau^K_t\right)\left(\frac{K^b_t}{L_t}\right)^2\right) / E_t \pi_{t+1}},$$

and increases with the share of debt in total financing. The entrepreneur's real return on capital is

$$r^K_t = \frac{z_t + Q_t(1-\delta)}{Q_{t-1}}, \tag{17}$$

where z_t is the real rental cost of capital. As in models with adjustment costs and incomplete depreciation, this return differs from the rental rate because it incorporates both current rental income and the valuation effect coming from changes in the price of capital.

Each period, a fraction $1-\nu$ of entrepreneurs exits and consumes its remaining net worth. New entrepreneurs enter with a transfer $\bar{g}$ from exiting entrepreneurs, so aggregate net worth evolves according to

$$N_t = \nu V_t + (1-\nu)\bar{g}, \tag{18}$$

where V_t denotes the net worth of surviving entrepreneurs. Following Gertler et al. (2007), we assume that lending is denominated in nominal terms. Surviving entrepreneurs therefore satisfy

$$V_t = (1-\tilde{\mu}) r^K_t Q_{t-1} K_{t-1} - i^L_{t-1} \frac{P_{t-1}}{P_t} L_{t-1}, \tag{19}$$

where $\tilde{\mu}$ captures the deadweight loss associated with imperfect credit markets and i^L_t

is the state-contingent nominal lending rate implied by the optimal financial contract. Combining equations (18) and (19) yields a dynamic equation for aggregate net worth. Entrepreneurs' balance sheets therefore depend both on asset-price movements and on ex-post borrowing costs, so unexpected changes in inflation affect net worth through the real value of nominal debt.

2.3 Households and Production

2.3.1 Households

There is a continuum of households that consume domestic and imported goods, supply labor, and allocate savings between bank deposits and foreign bonds.

The representative household derives utility from consumption C_t and disutility from labor h_t according to $u(C_t, h_t) = \ln C_t - \Psi \frac{h_t^{1+\phi}}{1+\phi}$. Consumption is a Cobb-Douglas bundle of home (C_t^H) and foreign (C_t^F) goods, such that $C_t \propto (C_t^H)^{\gamma}(C_t^F)^{1-\gamma}$, with corresponding price index $P_t = (P_t^H)^{\gamma}(P_t^F)^{1-\gamma}$.

Households can hold real deposits D_t and foreign nominal bonds B_t. To ensure stationarity, foreign bond holdings are subject to a small quadratic portfolio cost $\frac{\psi_B}{2}\left(\frac{S_t}{P_t}B_t\right)^2$. The budget constraint is

$$P_t C_t + P_t D_t + S_t B_t + \frac{\psi_B}{2} P_t \left(\frac{S_t}{P_t} B_t\right)^2 = i_{t-1}^D P_{t-1} D_{t-1} + i_{t-1}^* S_t B_{t-1} + P_t W_t h_t + P_t \Pi_t^R + P_t T_t, \tag{20}$$

where W_t is the real wage, Π_t^R denotes profits from intermediate-goods firms, and T_t collects lump-sum transfers, including seigniorage rebates.

Let $\Lambda_{t,t+k} = \beta^k C_t / C_{t+k}$ denote the stochastic discount factor. Household optimality implies

$$1 = E_t \Lambda_{t,t+1} \frac{i_t^D}{\pi_{t+1}}, \tag{21}$$

$$1 + \psi_B \frac{S_t}{P_t} B_t = E_t \left[\Lambda_{t,t+1} \frac{i_t^*}{\pi_{t+1}} \frac{S_{t+1}}{S_t} \right], \tag{22}$$

$$W_t = \Psi h_t^{\phi} C_t \,. \tag{23}$$

where $\pi_t = P_t / P_{t-1}$ is the gross inflation rate. These conditions link consumption growth to the return on deposits and determine the uncovered interest parity relationship modified by the portfolio adjustment cost.

2.3.2 Intermediate Goods

Intermediate goods producers buy labor from households and rent capital from entrepreneurs. They produce differentiated intermediate goods and operate in competitive input markets and monopolistically competitive output markets. The production technology of producer $i \in [0,1]$ is given by

$$y_t(i) = \xi_t^A K_{t-1}(i)^{\alpha} h_t(i)^{1-\alpha}, \tag{24}$$

where ξ_t^A denotes an aggregate technology shock, common across firms, which follows an AR(1) process.

Each producer chooses labor and capital inputs so as to minimize total costs for a given level of output. Cost minimization implies the standard optimal factor-demand condition

$$\frac{h_t(i) W_t}{z_t K_{t-1}(i)} = \frac{1-\alpha}{\alpha},$$

where W_t is the real wage and z_t is the real rental rate of capital. This condition equates the ratio of factor payments to the ratio of output elasticities in production.

Given the production function, real marginal costs are proportional to

$$mc_t \propto \frac{W_t^{1-\alpha} z_t^{\alpha}}{\xi_t^A}. \tag{25}$$

Hence, marginal costs increase with factor prices and fall with productivity. This link is important because movements in wages, rental rates, and technology feed directly into firms' pricing decisions and therefore into inflation dynamics.

2.3.3 Final Goods

A competitive final-good sector combines the continuum of domestic intermediate varieties into a single homogeneous domestic good. This final good is used both for household consumption and for investment. The aggregation technology is CES and takes the form

$$Y_t = \left(\int_0^1 y_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}},$$

where $\epsilon > 1$ denotes the elasticity of substitution across varieties.

Because final-good firms are competitive, they choose the bundle of intermediate inputs so as to minimize the cost of producing a given amount of final output. The

associated cost-minimization problem delivers the demand schedule for each variety $y_t(i) = \left(\frac{p_t^H(i)}{P_t^H}\right)^{-\epsilon} Y_t$, where $p_t^H(i)$ is the price charged by producer i and P_t^H is the domestic price index. Zero profits in the final-good sector then imply that the aggregate price index satisfies $P_t^H = \left(\int_0^1 p_t^H(i)^{1-\epsilon} di\right)^{\frac{1}{1-\epsilon}}$.

These objects are important because they determine the environment faced by intermediate producers when setting prices. We assume Calvo price rigidity, so the probability that a firm cannot reoptimize its price for k periods is given by θ^k. An intermediate firm that is allowed to reset its price at time t chooses p_t^* to maximize the expected discounted stream of real profits,

$$\max_{\{p_t^*\}} E_t \left[\sum_{k=0}^{\infty} \theta^k \Lambda_{t,t+k} \Pi_{t+k|t}^R(i) \right], \tag{26}$$

where

$$\Pi_{t+k|t}^R(i) = \frac{p_t^*}{P_{t+k}} y_{t+k|t}(i) - mc_{t+k|t}(i) y_{t+k|t}(i).$$

The corresponding optimal pricing condition is

$$E_t \left[\sum_{k=0}^{\infty} \theta^k \Lambda_{t,t+k} y_{t+k|t}(i) \left(\frac{p_t^*}{P_{t+k}} - \frac{\epsilon}{\epsilon - 1} mc_{t+k|t}(i) \right) \right] = 0. \tag{27}$$

Finally, imported goods are supplied competitively by the rest of the world, and their foreign-currency price is normalized to one. Hence, $P_t^F = S_t$.

2.4 Market Clearing in the Goods Market

The open-economy structure matters because macroprudential policy affects not only domestic credit conditions, but also the relative returns that shape portfolio choices, exchange-rate dynamics, and external demand. In particular, when reserve requirements compress deposit rates, they influence household saving behavior and can induce exchange-rate movements that are not replicated one-for-one by tighter capital requirements.

The model is closed by the resource constraint

$$Y_t = \gamma \frac{P_t}{P_t^H} (C_t + I_t + G_t + \Gamma_t) + \frac{S_t}{P_t^H} X_t$$

where X_t is exogenous foreign demand for domestic goods and Γ_t collects all the relevant

adjustment and agency costs in the model. The balance-of-payments identity is

$$S_t B_t = P_t^H Y_t - P_t(C_t + I_t + G_t + \Gamma_t) + (i_{t-1}^*) S_t B_{t-1} \ .$$

2.5 Government

We consider a government that implements both monetary and macroprudential policy, and therefore we refer to this authority as the central bank throughout the paper.[12] Our policy exercises have two components: a set of simple rules that governs the response of each policy instrument to macroeconomic and financial conditions, and a loss function that summarizes the central bank's stabilization objectives.

2.5.1 Policy instruments

Policy is implemented through three instruments: the interbank interest rate (i_t^{IB}), the reserve requirement (τ_t^R), and the capital requirement (τ_t^K). These instruments can potentially react to inflation, output, and the credit-to-output ratio through the following class of simple rules:

$$\hat{i}_t^{IB} = \phi_{\pi,i}\hat{\pi}_t + \phi_{Y,i}\hat{Y}_t + \phi_{l,i}\hat{l}_t \tag{28}$$

$$\tilde{\tau}_t^R = \phi_{Y,R}\hat{Y}_t + \phi_{l,R}\hat{l}_t \tag{29}$$

$$\tilde{\tau}_t^K = \phi_{Y,K}\hat{Y}_t + \phi_{l,K}\hat{l}_t \tag{30}$$

where $l_t \equiv L_t/Y_t$ is the credit-to-output ratio. The first subscript of each coefficient ϕ denotes the variable to which the instrument responds, and the second subscript identifies the instrument: i for the interbank rate, R for the reserve requirement, and K for the capital requirement. The interest rate rule is expressed as fractional deviations from their steady-state gross rates. Because macroprudential requirements are ratios, their deviations are expressed in level terms, so that $\tilde{\tau}_t^R \equiv \tau_t^R - \bar{\tau}^R$ and $\tilde{\tau}_t^K \equiv \tau_t^K - \bar{\tau}^K$, where $\bar{\tau}^R$ and $\bar{\tau}^K$ are the respective steady state values.[13]

The rules above allow us to represent different assignments of instruments to stabilization objectives. In the baseline arrangement, the interest rate rule follows a Taylor-type specification and does not respond directly to credit ($\phi_{l,i} = 0$), leaving credit stabilization to the macroprudential instruments. We restrict reserve and capital requirements not to respond to inflation ($\phi_{\pi,R} = \phi_{\pi,K} = 0$), since the purpose is to study them as macroprudential tools rather than as unconventional monetary policy instruments.

[12] Note that in some jurisdictions it may also encompass the role of a financial regulator.

[13] Consequently, $\tilde{\tau}_t^R = 0.01$ means a one percentage point increase in the reserve requirement, and $\tilde{\tau}_t^K = 0.01$ means a one percentage point increase in the capital requirement.

2.5.2 Objectives

The first objective we consider is a conventional stabilization in which the central bank cares about inflation and output volatility, measured as log-deviations from their steady state values.[14] For a given policy arrangement n, let Φ_n denote the vector of active rule coefficients. The core price-stability (PS) loss is

$$\mathcal{L}^{PS}(\Phi_n) = E\left[\hat{\pi}_t^2 + \lambda_Y \hat{Y}_t^2\right]. \tag{31}$$

The second objective adds a financial-stability component, measured by deviations of the credit-to-output ratio from its steady state. The core financial-stability (FS) loss is

$$\mathcal{L}^{FS}(\Phi_n) = E\left[\hat{\pi}_t^2 + \lambda_Y \hat{Y}_t^2 + \lambda_l \hat{l}_t^2\right]. \tag{32}$$

Here, λ_Y and λ_l govern the relative weights assigned to output and credit stabilization, respectively, relative to inflation stabilization.

When solving for optimal simple rules, the active coefficients in Φ_n are chosen to minimize an augmented loss that adds a penalty on movements in macroprudential requirements. The idea is to discipline the choice of optimized coefficients by ruling out policies that achieve stabilization only through implausible changes in the macroprudential instruments.[15] For objective $\omega \in \{PS, FS\}$, this augmented loss is

$$Q_n^{\omega}(\Phi_n) = \mathcal{L}^{\omega}(\Phi_n) + \lambda_\tau E\left[(\tilde{\tau}_t^R)^2 + (\tilde{\tau}_t^K)^2\right]. \tag{33}$$

The parameter λ_τ determines the weight placed on the volatility of macroprudential instruments. In the baseline, we set $\lambda_Y = 1$, $\lambda_l = 1$, and $\lambda_\tau = 0.10$, in line with the analysis in Hecker et al. (2024).[16] The credit weight λ_l applies only under the financial-stability objective. If a macroprudential requirement is not active under a given policy arrangement, its deviation from steady state is zero. In section 4.3 we further relax the assumption of $\lambda_l = 1$, in order to explore how the trade-off between macroeconomic stability and financial stability change when the macroprudential policies are active.

The inclusion of the credit-to-output ratio in $\mathcal{L}^{FS}$ is meant to capture financial sta-

[14] We refer to this as the "price-stability" objective, even if the central bank might be following a dual mandate (prices and output). This is in line with the usual concern for real activity in flexible inflation-targeting frameworks.

[15] This penalty also captures an implementation concern: frequent or large changes in macroprudential requirements may generate uncertainty for banks and borrowers and make policy less predictable. This is in line with the literature, where macroprudential authorities are often assumed to care not only about credit and output stabilization, but also about the volatility of the policy instrument itself (Hecker et al., 2024, Angelini et al., 2014, Ferrero et al., 2024). Unless otherwise stated, however, we rank and report policies using the corresponding core loss, $\mathcal{L}^{\omega}(\Phi_n)$.

[16] Appendix D reports robustness checks for alternative values of λ_τ.

bility concerns in a reduced form. Credit aggregates are commonly used as indicators of financial imbalances, and deviations of credit from trend have been shown to contain information about crisis risk, as in Borio and Lowe (2002), Borio and Drehmann (2009), and Drehmann et al. (2010). In our model, however, there is no explicit crisis state. Therefore, in this setting, credit stabilization focuses on the role that the macroprudential policies have on limiting credit fluctuations, while abstracting from measuring the benefits of preventing a financial crisis.

2.6 Shocks and Calibration

The model is driven by five fundamental shocks: a cost-push shock (ξ_t^{CP}), a technology shock (ξ_t^{A}), a government spending shock (G_t), a foreign interest rate shock (i_t^*), and a foreign export demand shock (X_t).[17] All shocks follow AR(1) processes. Their persistence and variances are reported in Table 1. We take these values from the estimated DSGE model in Christoffel et al. (2008). The remaining structural parameters are either standard in the literature or chosen to match relevant model targets, as summarized in Table 2. The steady state ratios and gross returns used in the log-linearized model are reported separately in Appendix B.

Among the parameters calibrated to steady state targets, the reserve requirement and entrepreneurial-finance blocks are important for the transmission of macroprudential policy. We follow Glocker and Towbin (2012) in setting the policy reserve requirement τ^R to 0.09 and the effective reserve ratio ς to 0.10. The remaining reserve-block parameters imply a quarterly steady state spread of 150 basis points between the interbank rate (i_t^{IB}) and the remuneration rate on reserves (i_t^R). For entrepreneurial finance, we set steady state leverage to two and calibrate the financial-contract parameters to generate an external finance premium of 70 basis points per quarter and an elasticity to leverage of $\eta = 0.05$, following Christensen and Dib (2008).

Finally, the calibration of the lending-unit block follows Gerali et al. (2010). This applies to the sensitivity of bank capital adjustment costs (κ), the debt-to-loans ratio, bank capital depreciation, and the target capital-to-loans ratio (τ^K). We set the steady state bank capital ratio to 0.11, above the regulatory minimum. This captures the fact that banks typically operate with capital buffers above required levels, reflecting prudential, reputational, and operational considerations (see Allen and Rai, 1996; Peura and Jokivuolle, 2004; Barth et al., 2013).

[17] Section 3 also considers two policy shocks, one to reserve requirements and one to capital requirements, to illustrate the transmission of the macroprudential instruments. In the optimal-policy exercises, however, these instruments are modeled as endogenous policy rules.

Table 1: Calibration of the Shocks

ρ	σ^2	Description
0.89	1.13	Technology Shock
0.40	0.14	Cost-Push Shock
0.86	4.63	Government Spending Shock
0.88	0.43	Foreign Interest Rate Shock
0.80	5.01	Export Demand Shock

Note: The table reports the persistence parameter ρ and innovation variance σ^2 of the five AR(1) shock processes. Values are taken from Christoffel et al. (2008).

Table 2: Calibration

Param.	Value	Description
δ	0.025	Depreciation Rate of Capital
β	0.985	Discount Factor
α	0.33	Capital Share in Production
ϕ	3.00	Inverse of Frisch Labor Supply Elasticity
θ	0.75	Degree of Price Stickiness
ν	0.97	Survival Rate of Entrepreneurs
χ	0.25	Capital Adjustment Costs
η	0.05	Elasticity of External Finance Premium
ψ_B	0.02	Adjustment Costs for Net Foreign Assets
γ	0.75	Share of Domestically Produced Goods
δ^b	0.1049	Bank Capital Depreciation
κ	3.63	Sensitivity to Bank Capital Cost

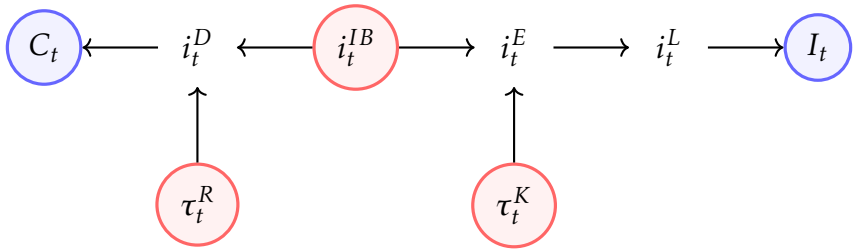
Note: The table reports the main structural parameters of the model. Steady-state values and additional calibration targets are reported in Appendix B.

3 Impact of Macroprudential Policies

In this section, we examine the transmission mechanisms and macroeconomic effects of discretionary changes in reserve and capital requirements. This is a useful step for understanding how the two policy instruments operate before asking how they should be used by the central bank. We model both instruments as exogenous AR(1) processes with autocorrelation 0.5, and abstract from any systematic policy component. Monetary policy is kept as simple as possible throughout, following a Taylor rule that responds only to inflation deviations with coefficient $\phi_{\pi,i} = 1.5$.[18] We pay particular attention to the role of the financial accelerator in amplifying the effects of each instrument, and

[18] All other coefficients in the Taylor rule are set to zero.

Figure 2: Transmission Mechanism of Macroprudential Policy



Note: Variables in red denote the monetary and macroprudential policy instruments. i_t^D, i_t^E, and i_t^L are the deposit, effective lending, and state-contingent lending rates, respectively. C_t and I_t denote aggregate consumption and investment. Arrows indicate the primary direction of influence between variables, but in general equilibrium all variables are jointly determined.

return to the role of nominal frictions when comparing their aggregate effects.[19]

Figure 2 provides a simple illustration of the transmission channels through which each macroprudential tool operates. Both instruments are expected to affect the spread between deposit and lending rates, balance-sheet composition, and aggregate variables, although through different margins. Reserve requirements directly influence the distribution of deposits and reserves on banks' balance sheets, and therefore tend to have a more immediate impact on deposit rates. Households respond by adjusting their savings decisions, which in turn affects aggregate consumption. Capital requirements, by contrast, act more directly on the stock of loans and consequently on lending rates. Entrepreneurs, who rely on bank credit to fund their projects, respond by adjusting investment. Finally, the monetary policy rate links both sides of the balance sheet, influencing deposit and lending rates simultaneously.

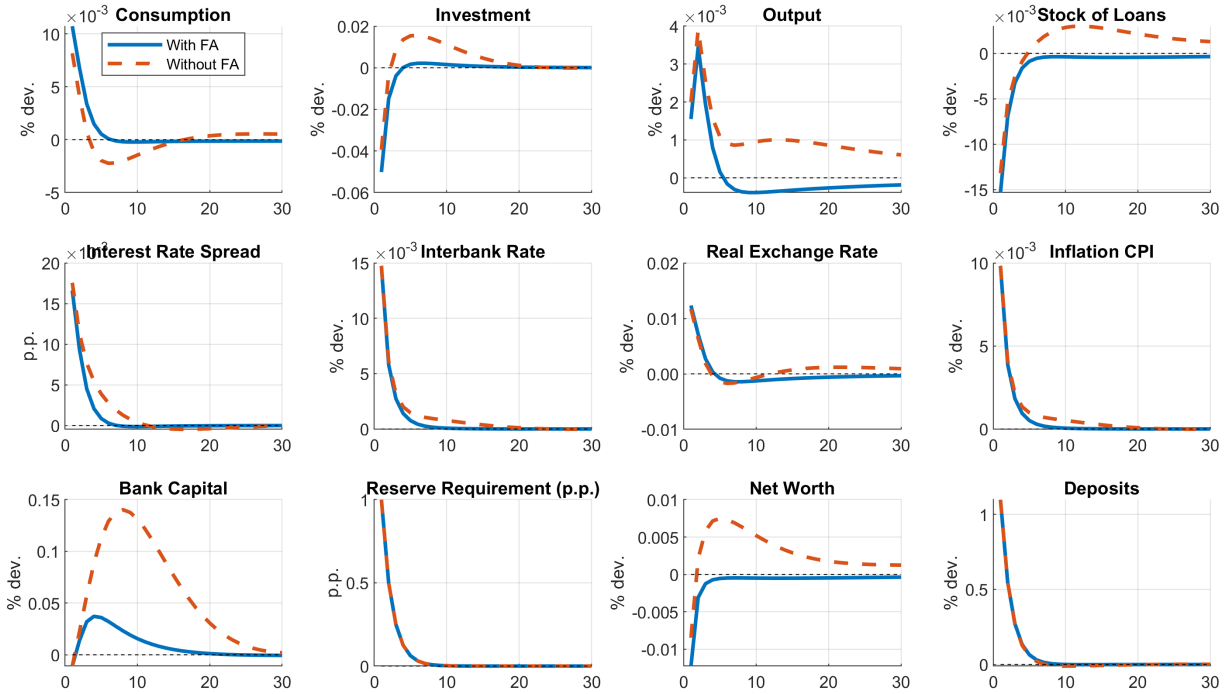
3.1 Transmission to Credit and Economic Activity

Figure 3 shows the impulse responses to a 1 p.p. increase in reserve requirements. The increase acts as a tax on bank intermediation, driving a wedge between deposit and lending rates. From a partial-equilibrium perspective, wider spreads discourage savings and boost consumption, while the fall in deposit rates triggers an exchange rate depreciation through uncovered interest parity, boosting exports. The net effect on output is therefore ambiguous: for our calibration, the consumption and export channels initially dominate the decline in investment, so output rises on impact before turning contractionary.

A similar exercise for capital requirements is reported in Figure 4. In this case, the mechanism works primarily through the lending side: higher requirements raise the marginal cost of lending relative to bank capital and induce banks to reduce loans, com-

[19] Results under alternative degrees of price stickiness are reported in Figures 7 and 8 in Appendix C.

Figure 3: Reserve Requirement Shock and the Financial Accelerator

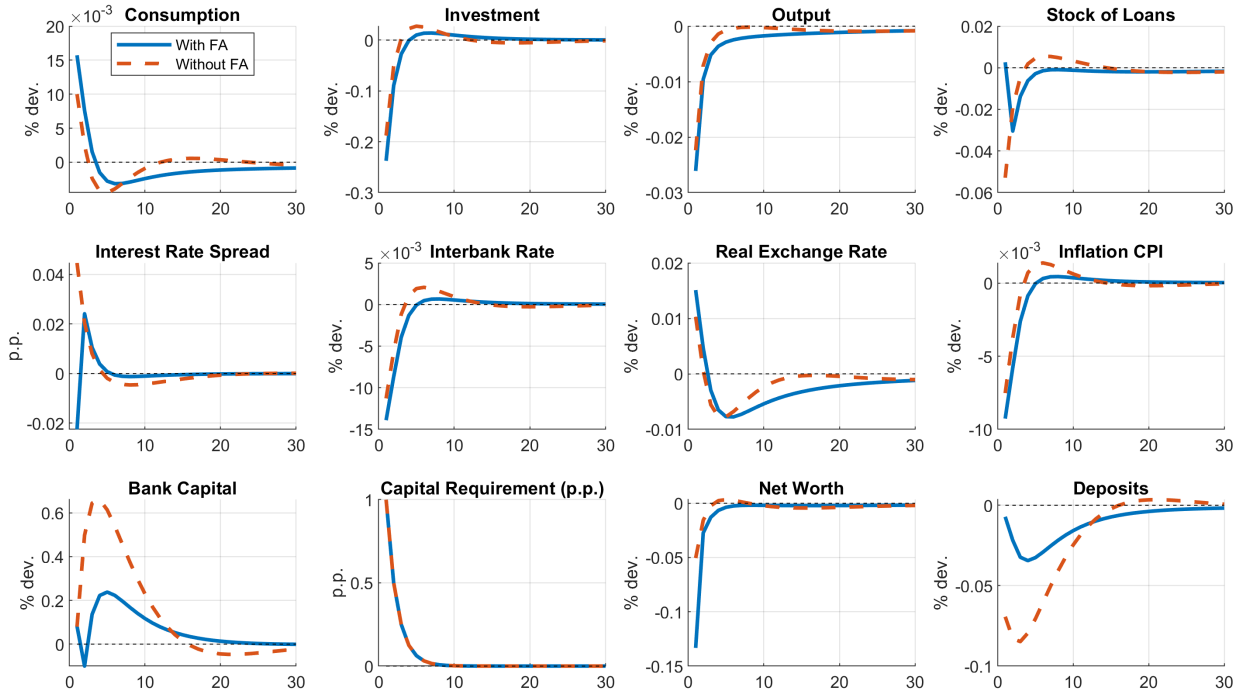


Note: The figure reports quarterly impulse responses to a 1 p.p. increase in reserve requirements, considering scenarios with and without the financial accelerator mechanism. Monetary policy is specified by an interest rate rule for the interbank interest rate as defined in Section 2 with $\phi_{\pi,i} = 1.5$ and the other coefficients equal 0. The y-axis denotes the deviation from the steady state either in percent or in percentage points.

pressing credit supply and pushing lending rates up. The deposit rate also falls, generating similar exchange rate and export dynamics as in the reserve requirement case, but the decline in investment is large enough to dominate. As a result, output falls on impact and remains contractionary throughout.

The financial accelerator amplifies the transmission of both instruments, though it plays a more prominent role in the case of reserve requirements. In both cases, movements in the external finance premium make entrepreneurial net worth and investment more sensitive to fluctuations in the interbank rate, resulting in a sharper and more persistent contraction in economic activity. For capital requirements, the accelerator mainly affects the magnitude of the decline in lending and interbank borrowing, leaving the qualitative results largely unchanged. For reserve requirements, by contrast, the amplification is stronger. In that case the financial accelerator reinforces the investment channel enough to eventually dominate the initial consumption and export boost, making the output contraction more severe and persistent than in the baseline case.

Figure 4: Capital Requirement Shock and the Financial Accelerator



Note: The figure reports quarterly impulse responses to a 1 p.p. increase in capital requirements, considering scenarios with and without the financial accelerator mechanism. Monetary policy is specified by an interest rate rule for the interbank interest rate as defined in Section 2 with $\phi_{\pi,i} = 1.5$ and the other coefficients equal 0. The y-axis denotes the deviation from the steady state either in percent or in percentage points.

3.2 Inflation and Monetary Policy

The most striking difference between the two instruments lies in their effects on inflation. In the case of capital requirements, inflation falls alongside output, as shown in Figure 4, consistent with a standard demand contraction. For reserve requirements, Figure 3 shows that both output and inflation rise on impact, as the consumption and export channels initially dominate the decline in investment. In this context, we see that the increase in the cost of bank intermediation ultimately puts upward pressure on prices. As these forces dissipate, both output and inflation gradually return to steady state. The two instruments can therefore generate opposite inflation dynamics, despite both tightening credit conditions.

This asymmetry has direct implications for monetary policy. A tightening of capital requirements reduces credit, output, and inflation. The fall in inflation induces the central bank to cut the interbank rate, so monetary policy partly offsets the contraction in economic activity while capital requirements lean against credit. In this case, price-stability and financial-stability motives are broadly aligned. Reserve requirements generate a more difficult policy mix. A tightening of reserve requirements also reduces

credit, but it can put upward pressure on inflation. The monetary authority then raises the interbank rate to stabilize prices, which further tightens financial conditions and amplifies the decline in credit. Thus, the use of reserve requirements as a countercyclical tool may create a trade-off between containing credit and stabilizing inflation, a tension that will be central when discussing optimal policies in Section 4.

The quantitative strength of these effects depends on the calibration, but price stickiness is a key determinant of the monetary policy response. With more flexible prices, reserve requirements generate a stronger increase in inflation, inducing a larger rise in the policy rate and reinforcing the contraction in credit. With stickier prices, cost pressures pass through more slowly, so the monetary policy response is smaller. Price stickiness therefore softens the conflict between credit containment and inflation stabilization in the case of reserve requirements, while also smoothing the fall in inflation after capital requirement shocks.[20]

4 Optimal Policy Rules

After illustrating the main transmission mechanisms of macroprudential policy, we turn to study the optimal policy rules a central bank can implement. We compare alternative policy arrangements that differ in which instruments are active and in the variables to which they respond, following the structure in Section 2.5. For each arrangement, the active coefficients in Φ_n are chosen to minimize the augmented objective in equation (33). We then report the policy performance using the corresponding core loss, $\mathcal{L}^{PS}$ or $\mathcal{L}^{FS}$, as described in Section 2.5. In this way, the optimization penalizes excessively volatile movements in macroprudential requirements, while the ranking of policy rules remains tied to the underlying stabilization objective.

4.1 Price-stability Objective

We first consider the price-stability objective, under which the central bank cares about inflation and output volatility but does not directly react to credit fluctuations. Table 3 reports the optimized coefficients and the resulting core loss for three policy rules. Policy A(I) is the benchmark: monetary policy follows a Taylor-type rule, responding to inflation and output, while macroprudential requirements are inactive.

Policies A(II) and A(III) add one macroprudential instrument at a time. In Policy A(II), reserve requirements respond to the credit-to-output gap. In Policy A(III), capital requirements respond to the same variable. In both cases, the credit-to-output gap enters

[20] Results under alternative degrees of price stickiness are reported in Figures 7 and 8 in Appendix C.

Table 3: Optimal Policy Rules under a Price-stability Objective

	Policy A(I)	Policy A(II)	Policy A(III)
interest rate rule	$\hat{\pi}_t, \hat{Y}_t$	$\hat{\pi}_t, \hat{Y}_t$	$\hat{\pi}_t, \hat{Y}_t$
Macroprudential rule	—	$\tau_t^R \to \hat{l}_t$	$\tau_t^K \to \hat{l}_t$
$\phi_{\pi,i}$	2.532	2.534	2.541
$\phi_{Y,i}$	0.653	0.653	0.657
$\phi_{l,R}$	—	0.019	—
$\phi_{l,K}$	—	—	0.004
$10^4 \times \mathcal{L}^{PS}(\Phi_n)$	7.6156	7.6151	7.6156
$\Delta\mathcal{L}^{PS}$ relative to benchmark	—	−0.01%	−0.00%

Note: The table reports optimized coefficients for the policy arrangements in block A. Policy A(I) is the benchmark in which the interbank rate responds to inflation and output. Policy A(II) adds a reserve requirement response to the credit-to-output gap. Policy A(III) adds a capital requirement response to the credit-to-output gap. The loss is the core price-stability loss in equation (31). Lower values indicate better stabilization. A dash denotes that the coefficient is not active in the corresponding policy arrangement.

the policy rule as an indicator of macroeconomic and financial conditions, not because credit stabilization is part of the central bank's objective itself.

Our results show that adding macroprudential responses to the credit-to-output gap produces virtually no additional stabilization gains under a price-stability objective. The reserve requirement rule reduces the core loss by 0.01%, while the capital requirement rule does not reduce it. This is informative. When the central bank cares only about inflation and output, credit matters only to the extent that it helps stabilize these two variables. In the model, most of the relevant information about the economic cycle is already captured by the Taylor rule through price and output volatility. Movements in the credit-to-output gap also reflect financial and compositional forces that do not necessarily call for an additional policy response. As a result, under a price-stability objective, macroprudential rules play at most a supporting role for the central bank.

4.2 Financial-stability Objective

We now consider the financial-stability objective, in which the central bank cares not only about inflation and output volatility, but also about stabilizing the credit cycle. The results are reported in Tables 4 and 5. Policy block B keeps the benchmark monetary policy rule simple, with the interbank rate responding to inflation and output. Policy block C instead moves toward a clearer separation of tasks, with monetary policy assigned to inflation and the macroprudential instruments assigned to output and credit. Within this block, we consider both a broad joint rule and stricter assignments in which each

Table 4: Optimal Policy Rules under a Financial-stability Objective: Policy Block B

	Policy B(I)	Policy B(II)	Policy B(III)	Policy B(IV)
Interest rate rule	$\hat{\pi}_t, \hat{Y}_t$	$\hat{\pi}_t, \hat{Y}_t, \hat{l}_t$	$\hat{\pi}_t, \hat{Y}_t$	$\hat{\pi}_t, \hat{Y}_t$
RR rule	—	—	$\hat{Y}_t, \hat{l}_t$	—
CR rule	—	—	—	$\hat{Y}_t, \hat{l}_t$
$\phi_{\pi,i}$	1.487	1.199	1.501	1.619
$\phi_{Y,i}$	0.200	0.017	0.202	0.252
$\phi_{l,i}$	—	−0.111	—	—
$\phi_{Y,R}$	—	—	0.450	—
$\phi_{l,R}$	—	—	0.638	—
$\phi_{Y,K}$	—	—	—	0.712
$\phi_{l,K}$	—	—	—	0.994
$10^4 \times \mathcal{L}^{FS}(\Phi_n)$	13.62	13.30	13.22	12.72
$\Delta\mathcal{L}^{FS}$ relative to benchmark	—	−2.37%	−2.96%	−6.62%

Note: The table reports optimized coefficients for policy block B. Policy B(I) is the benchmark in which the interbank rate responds to inflation and output. Policy B(II) adds a monetary policy response to the credit-to-output gap. Policy B(III) adds a reserve requirement rule. Policy B(IV) adds a capital requirement rule. The loss is the core financial-stability loss in equation (32). Lower values indicate better stabilization. A dash denotes that the coefficient is not active in the corresponding policy arrangement.

macroprudential instrument responds to only one variable.

Policy B(I) in Table 4 presents our baseline scenario where only the coefficients $\phi_{\pi,i}$ and $\phi_{Y,i}$ are active. Compared to the analogous rule under the price-stability objective, the loss is now higher because the central bank has an additional objective, but the same number of instruments. This first result suggests that there is space for macroprudential tools to improve stabilization once credit fluctuations enter the objective directly.

Policy B(II) extends the simple interest rate rule by allowing the policy rate to respond also to the credit-to-output gap. This augmented Taylor rule delivers a modest gain, reducing the core financial-stability loss by 2.37% relative to the benchmark. This suggests that in this environment, giving the monetary policy rate a macroprudential role can help but only marginally, once it is already being used to stabilize inflation and output.

In this context, introducing a proper macroprudential instrument delivers larger gains. Policies B(III) and B(IV) add a reserve requirement rule and a capital requirement rule, respectively, allowing the macroprudential tool to respond to output and credit volatility.[21] This lets the additional instrument operate directly through banks' balance sheets,

[21] The Basel guidance gives a prominent role to the credit-to-GDP gap, but not as a mechanical trigger. In practice, authorities often complement it with broader indicators of the cycle. Peru is a clear example, since its countercyclical capital framework uses GDP-growth activation criteria. In Europe, frameworks

Table 5: Optimal Policy Rules under a Financial-stability Objective: Policy Block C

	Policy C(I)	Policy C(II)	Policy C(III)	Policy C(IV)
Interest rate rule	$\hat{\pi}_t$	$\hat{\pi}_t$	$\hat{\pi}_t$	$\hat{\pi}_t$
RR rule	—	$\hat{Y}_t, \hat{l}_t$	$\hat{l}_t$	$\hat{Y}_t$
CR rule	—	$\hat{Y}_t, \hat{l}_t$	$\hat{Y}_t$	$\hat{l}_t$
$\phi_{\pi,i}$	1.355	1.435	1.369	1.404
$\phi_{Y,R}$	—	0.336	—	0.116
$\phi_{l,R}$	—	0.633	0.480	—
$\phi_{Y,K}$	—	0.726	0.423	—
$\phi_{l,K}$	—	0.842	—	0.509
$10^4 \times \mathcal{L}^{FS}(\Phi_n)$	14.71	13.51	14.12	14.38
$\Delta\mathcal{L}^{FS}$ relative to benchmark	—	−8.20%	−4.03%	−2.25%

Note: The table reports optimized coefficients for policy block C. Policy C(I) is the benchmark in which the interbank rate responds only to inflation. Policies C(II)–C(IV) assign reserve requirements and capital requirements to output and/or credit stabilization. The loss is the core financial-stability loss in equation (32). Lower values indicate better stabilization. A dash denotes that the coefficient is not active in the corresponding policy arrangement.

instead of forcing the monetary policy rate to absorb the credit-stabilization role. The reserve requirement rule reduces the loss by 2.96%, while the capital requirement rule reduces it by 6.62%. Thus, once financial stability is part of the objective, macroprudential rules improve performance more clearly than an additional credit response in the interest rate rule.

The comparison between Policies B(III) and B(IV) also provides a first view of how the two macroprudential instruments perform as substitutes. In this setup, where both instruments are allowed to respond to both output and credit, the lower loss under Policy B(IV) indicates that capital requirements are the more effective single macroprudential tool. This comparison does not isolate the credit margin, however. Section 6 returns to the substitution question by assigning each instrument only to credit fluctuations and comparing their relative performance against the different shocks in the model.

Table 5 turns from comparing reserve and capital requirements as alternative macroprudential instruments to asking how they should be combined when both are available. This exercise is useful because in practice, a clearer separation of tasks may improve the communication from the central bank to firms and households, and reduce uncertainty about future policy changes, even though these benefits are not explicitly captured by the model. This motivates considering more restrictive policy arrangements in which each instrument is given a clearly defined role. Policy C(I) is the benchmark in which

such as those of Spain and the Netherlands also monitor real-activity indicators, including the output gap or GDP growth, when assessing cyclical systemic risk. This motivates allowing macroprudential rules to respond to output as well as credit.

monetary policy responds only to inflation. Policy C(II) allows both macroprudential instruments to respond to output and credit, and delivers the largest gain within this block, reducing the core financial-stability loss by 8.20%. This broad assignment gives the macroprudential instruments more flexibility, but it is also less parsimonious and may be harder to communicate as a policy rule.

Policies C(III) and C(IV) impose cleaner assignments. In Policy C(III), reserve requirements respond to the credit-to-output gap and capital requirements respond to output. In Policy C(IV), the rule is reversed. Our results favor Policy C(III), which reduces the loss by 4.03%, compared with 2.25% under Policy C(IV). The difference is not large, so we interpret this ranking with caution. Still, the result gives a first indication that the two instruments do not need to be simple substitutes. Even when capital requirements are the stronger direct credit tool, reserve requirements can complement them by taking on the credit-stabilization role, leaving capital requirements to respond to output. Section 6 examines whether this preferred assignment changes with the source of fluctuations in the economy.

Overall, the financial-stability objective strengthens the case for assigning financial stabilization to macroprudential tools rather than to the interest rate rule. Block B shows that a dedicated macroprudential instrument performs better than extending the monetary policy rule to credit fluctuations and, among the two macroprudential tools, capital requirements deliver the larger stabilization gain. Block C then shifts the focus to their joint use, showing that under a strict separation of tasks, the preferred assignment gives reserve requirements the response to credit and capital requirements the response to output. This gives us a first hint of how the two instruments can complement each other through specialization. The next subsection asks whether these policy rankings remain robust as the central bank places progressively more weight on credit stabilization.

4.3 Stabilization Frontier

The previous exercises showed that the relative appeal of reserve and capital requirements depends on the policy role assigned to each instrument. We now ask how these conclusions change as the central bank becomes increasingly more concerned with financial stability. To do so, we progressively increase the weight (λ_l) placed on stabilizing the credit-to-output gap relative to inflation and output, re-optimizing the policy rules at each step. This allows us to trace how the attainable balance between macroeconomic and financial stability changes across alternative policy arrangements.

For each optimized rule, Figure 5 plots credit-to-output volatility against an index of

macroeconomic volatility. The two coordinates are

$$\mathcal{V}_n^l(\lambda_l) \equiv 100\sqrt{\text{var}(\hat{l}_t)},$$
$$\mathcal{V}_n^M(\lambda_l) \equiv 100\sqrt{\text{var}(\hat{\pi}_t) + \text{var}(\hat{Y}_t)},$$

where n denotes the policy regime. A greater stabilization is achieved when the value of either indicator falls, making the southwest corner the most desired location for a central bank. As we increase λ_l, the coordinates resulting from the optimized policy should move farther to the left, indicating lower credit volatility. This yields a "stabilization frontier" in which we can compare the trade-offs between macroeconomic and financial stability objectives that the policymaker faces. Note that the filled markers in Figure 5 denote the baseline value $\lambda_l = 1$.[22]

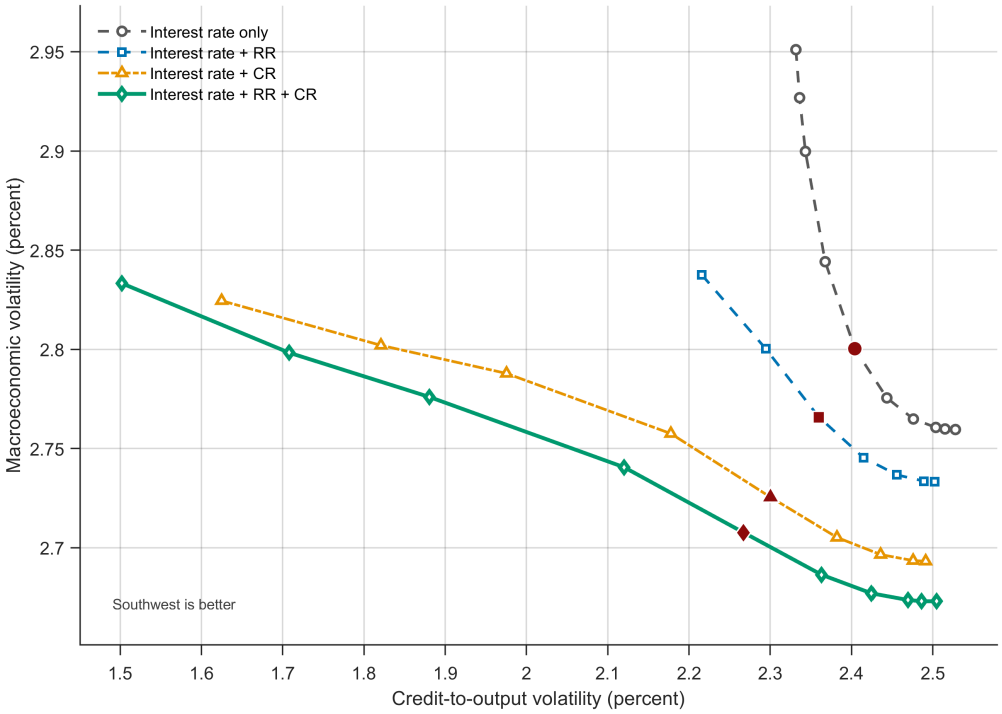
We start with the simplest case, in which the central bank is constrained and can only use monetary policy for macroeconomic and financial stability (Policy B(I)). We can see that varying the importance of financial stability does little to move the volatility coordinates along the horizontal axis, showing the limited capacity to stabilize credit. More importantly, increasing the importance of credit in the loss function requires stronger sacrifices in macroeconomic stability, tracing the steep slope on the left side of the gray line.

Once the central bank can rely on macroprudential policies to stabilize the cycle, the trade-off improves substantially. Both reserve requirements (blue frontier) and capital requirements (yellow frontier) shift the attainable trade-off to the left. This means that either instrument can help reduce credit volatility without requiring the same deterioration in macroeconomic stability. This suggests that the two instruments can be viewed as imperfect substitutes, with capital requirements consistently delivering a more favorable trade-off than reserve requirements across the range of λ_l.

When both macroprudential instruments are available, the green frontier lies further to the left than the frontiers obtained with either instrument alone. This shows the additional stabilization gains from being able to act on both funding and lending margins. At the same time, this does not by itself establish complementarity, since the policymaker also has access to a broader set of instruments. We therefore assess complementarity more directly through the strict division of tasks in Section 4.2, and later, in Section 6.

[22] See Appendix E for details on the construction of the stabilization frontiers and the policy rules underlying each line.

Figure 5: Stabilization frontiers under alternative policy arrangements



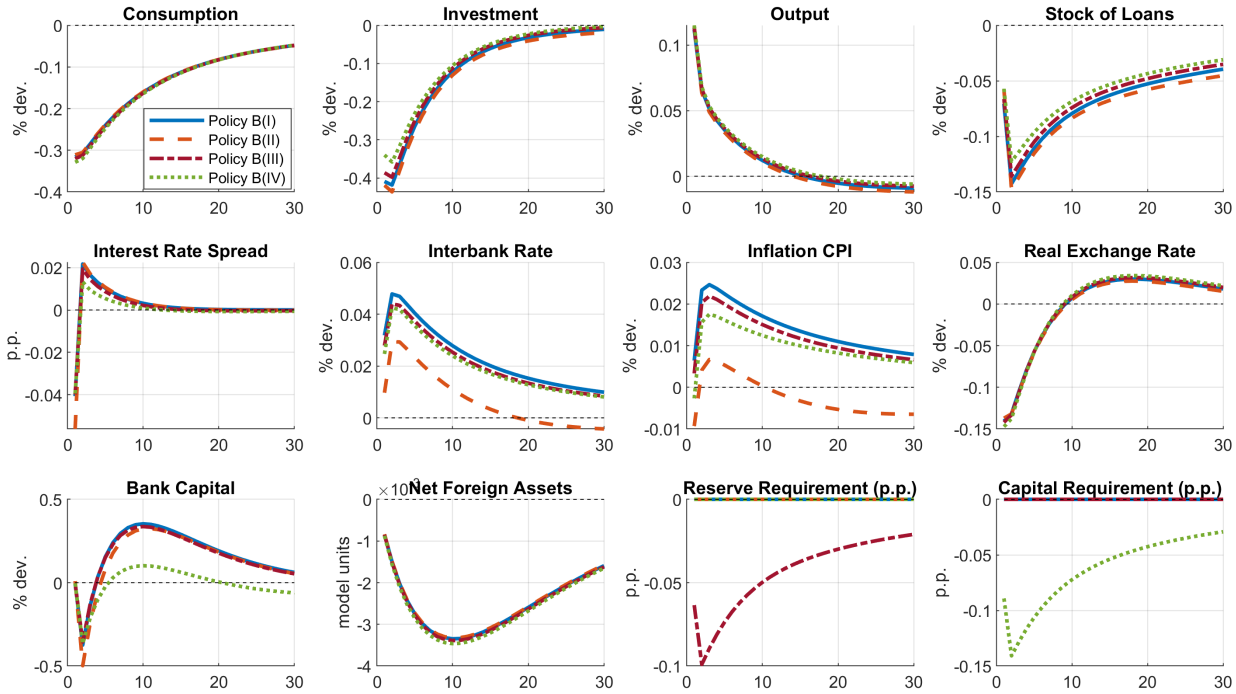
Note: The horizontal axis reports the standard deviation of the credit-to-output gap, $\mathcal{V}_n^l(\lambda_l) = 100\,\mathrm{sd}(\hat{l}_t)$, while the vertical axis reports macroeconomic volatility, $\mathcal{V}_n^M(\lambda_l) = 100\sqrt{\mathrm{var}(\hat{\pi}_t) + \mathrm{var}(\hat{Y}_t)}$. Each marker is obtained by re-optimizing the active policy coefficients for a different value of the financial-stability weight λ_l, holding $\lambda_Y = 1$ and $\lambda_\tau = 0.10$. "Interest rate only" uses the instrument set of Policy B(I). "Interest rate + RR" uses Policy B(III). "Interest rate + CR" uses Policy B(IV). "Interest rate + RR + CR" is an extension of Policies B(III) and B(IV) in which both reserve and capital requirements respond to output and the credit-to-output gap. The displayed solutions are drawn from a grid spanning $\lambda_l = 0$ to $\lambda_l = 8$. Filled markers identify the baseline financial-stability weight, $\lambda_l = 1$. Empty markers correspond to alternative values of λ_l. Along each frontier, higher values of λ_l are located toward the left, reflecting the greater weight placed on reducing credit-to-output volatility, while lower values are located toward the right.

5 Monetary and Macroprudential Trade-offs

In the previous section we compared alternative policy arrangements and showed how macroprudential instruments can improve the trade-off between macroeconomic and financial stability. We now turn to the interaction between monetary and macroprudential policy. The value of having separate instruments should become particularly clear when macroeconomic and financial conditions call for policy responses in different directions.

To illustrate this point, we consider a government-spending shock. In our model, higher government spending raises output and inflation, but crowds out private investment. This in turn represents a lower demand for credit, and therefore the stock of loans drops, creating an interesting policy trade-off. Stabilizing inflation and output calls for a tighter monetary policy response, while limiting the contraction in credit requires a

Figure 6: Government shock under Policy Block B and a Financial Stability Objective



Note: The figure reports quarterly impulse responses to a 1-std deviation increase in government expenditure, considering a scenario with a financial accelerator mechanism and different policy rules, as described in the legend. The y-axis denotes the deviation from the steady state either in percent or in percentage points.

more accommodative response on the macroprudential side.

Figure 6 compares the response to the government-spending shock under the policy arrangements in Block B and a financial-stability objective. We first consider Policies B(I) and B(II), where the interest rate is the only active policy instrument. Policy B(I) follows a standard Taylor rule, while Policy B(II) augments it with a response to the credit-to-output gap.

Under Policy B(I), the fiscal expansion raises both output and inflation, leading the central bank to increase the interbank rate. Under Policy B(II), the optimized rule places considerably less weight on the expansion in economic activity. In particular, the coefficient on output falls from 0.200 to 0.017, while the coefficient on inflation declines from 1.487 to 1.199. As a result, the interest-rate response is significantly softer. The different policy rule also changes the equilibrium path of private demand and financial conditions, resulting in a lower path of marginal costs and therefore a considerably smaller

increase in inflation.[23]

Policies B(III) and B(IV) instead, by introducing a dedicated macroprudential instrument, allow the interest rate to return to its traditional role of responding to inflation and output. The policy responses now move in different directions. The interbank rate increases in response to the fiscal expansion, while reserve or capital requirements, in policies B(III) and B(IV) respectively, are relaxed as credit contracts. This division of tasks softens the decline in lending and investment relative to the cases in which monetary policy acts alone, while also containing the increase in inflation.

Within this common pattern, the two macroprudential instruments generate only modestly different aggregate responses. Inflation rises somewhat less under the capital requirement rule than under the reserve requirement rule, and the interbank rate follows a similar pattern, although the difference is smaller. Lending and investment also contract slightly less under Policy B(IV). These differences are relatively limited, however, and the main feature shared by both policy arrangements is the separation of tasks between monetary and macroprudential policy.

The government-spending shock provides a particularly clear illustration because output and credit move in opposite directions. This is not the case for all disturbances. Following a positive technology shock, for example, both output and credit expand while inflation falls, so the main tension arises between price stabilization and the stabilization of real activity and credit. A foreign interest-rate shock also combines several competing channels, as higher foreign returns weaken domestic aggregate demand and credit, while the adjustment in the exchange rate raises inflation. The corresponding impulse responses are reported in Appendix C.

This exercise therefore illustrates why monetary and macroprudential policies may optimally move in different directions. When a shock raises inflation and output while depressing credit, monetary policy is called to tighten, whereas macroprudential policy can ease to smooth the contraction in lending, and therefore, in investment. A dedicated macroprudential instrument therefore allows credit fluctuations to be addressed without forcing the interest rate to accommodate conflicting objectives. Having established the value of separating monetary and macroprudential tasks, the next section turns to the division of tasks within macroprudential policy itself.

[23] In the absence of a cost-push innovation, domestic inflation satisfies the forward-looking Phillips curve

$$\hat{\pi}_t^d = \kappa_p E_t \sum_{j=0}^{\infty} \beta^j \hat{mc}_{t+j}, \qquad \kappa_p \equiv \frac{(1-\theta)(1-\theta\beta)}{\theta}.$$

Inflation therefore depends on the expected path of marginal costs, which itself changes with the policy rule.

6 Macroprudential Assignments across Shocks

The aggregate results in Section 4 provide two initial conclusions about the relationship between reserve and capital requirements. Policy Block B in Table 4 shows that capital requirements perform better when the two instruments are used as alternative macroprudential tools. Policy Block C in Table 5 then shows that, conditional on both instruments being available and under a strict separation of tasks, the preferred unconditional assignment gives reserve requirements the response to credit and capital requirements the response to output. These results, however, average across all shocks in the model. We now ask whether the relative performance of the two instruments depends on the source of fluctuations.[24]

We proceed in two steps. First, we sharpen the substitution comparison in Policy Block B by assigning either reserve or capital requirements directly to the credit-to-output gap and comparing their relative performance across the different shocks in the model. Second, we revisit the joint use of the two instruments considered in Policy Block C by assigning one instrument to output and the other to credit, and ask whether the preferred division of tasks changes with the source of fluctuations.

For each shock $u \in \mathcal{U} \equiv \left\{\xi_t^{CP}, \xi_t^{A}, G_t, i_t^*, X_t\right\}$, we solve the optimal rule problem with only that innovation active. Let $\mathcal{L}_u^{FS}(\Phi_n)$ denote the shock-specific core financial-stability loss under policy arrangement n:

$$\mathcal{L}_u^{FS}(\Phi_n) = E_u\left[\hat{\pi}_t^2 + \lambda_Y \hat{Y}_t^2 + \lambda_l \hat{l}_t^2\right]. \tag{34}$$

As in the unconditional exercises, the numerical optimization also includes the macroprudential-instrument volatility penalty. Policy arrangements are nevertheless compared using the core loss in equation (34). For policy arrangement n and benchmark arrangement b, the reported gain is

$$G_{n,u}^{FS} = 100 \times \frac{\mathcal{L}_u^{FS}(\Phi_b) - \mathcal{L}_u^{FS}(\Phi_n)}{\mathcal{L}_u^{FS}(\Phi_b)}. \tag{35}$$

The resulting percentage reductions in the core loss, relative to the relevant benchmark for each shock, are reported below. Additional details on the shock-specific optimization are reported in Appendix F.

Panel A of Table 6 sharpens the substitution comparison between the two instruments.[25] When reserve and capital requirements are assigned the same task—responding

[24] This exercise is not meant to imply that policymakers should condition policy rules on the realization of individual structural shocks, nor that such shocks are directly observable in real time. It is used to understand which shocks drive the unconditional policy rankings.

[25] This exercise is the shock-specific counterpart of Policy Block B in Table 4. The transmission mecha-

Table 6: Shock-specific Instrument-assignment Gains

Panel A: Which tool is better at leaning against credit fluctuations?			
Shock	RR $\rightarrow \hat{l}_t$	CR $\rightarrow \hat{l}_t$	Preferred tool
Technology/productivity, ξ_t^A	0.17	0.50	CR
Cost-push, ξ_t^{CP}	0.27	0.66	CR
Government spending, G_t	8.88	19.37	CR
Export demand, X_t	1.05	1.88	CR
Foreign interest rate, i_t^*	3.59	8.71	CR

Panel B: When macroprudential tools split tasks, which assignment is preferred?			
Shock	RR $\rightarrow \hat{l}_t$, CR $\rightarrow \hat{Y}_t$	RR $\rightarrow \hat{Y}_t$, CR $\rightarrow \hat{l}_t$	Preferred assignment
Technology/productivity, ξ_t^A	1.46	0.26	RR $\rightarrow \hat{l}_t$, CR $\rightarrow \hat{Y}_t$
Cost-push, ξ_t^{CP}	9.03	3.01	RR $\rightarrow \hat{l}_t$, CR $\rightarrow \hat{Y}_t$
Government spending, G_t	7.64	15.59	RR $\rightarrow \hat{Y}_t$, CR $\rightarrow \hat{l}_t$
Export demand, X_t	2.56	1.35	RR $\rightarrow \hat{l}_t$, CR $\rightarrow \hat{Y}_t$
Foreign interest rate, i_t^*	7.00	8.68	RR $\rightarrow \hat{Y}_t$, CR $\rightarrow \hat{l}_t$

Note: Entries are percentage reductions in the shock-specific core financial-stability loss $\mathcal{L}_u^{FS}$ relative to the benchmark in the same panel and shock. The first column reports the shock name and the corresponding model notation. When reserve or capital requirements are active, the OSR objective includes a macroprudential-instrument volatility penalty with weight $\lambda_\tau = 0.10$, but assignments are ranked using the core loss. RR denotes reserve requirements, CR denotes capital requirements.

only to the credit-to-output gap—capital requirements deliver a larger reduction in the core loss for every shock. In this context, the two instruments can therefore be viewed as imperfect substitutes, with capital requirements as a stronger direct tool for credit stabilization. Capital requirements operate directly through banks' lending capacity and loan spreads, whereas reserve requirements affect credit more indirectly through liquidity and funding conditions. The quantitative advantage is relatively modest for technology, cost-push, and export-demand shocks, but substantially larger for government-spending and foreign-interest-rate shocks. The aggregate ranking is therefore not driven by a single shock, although its strength varies considerably with the source of fluctuations.

Naturally, leaning against credit does not necessarily imply a macroprudential tightening. The rule tightens when credit-to-output is high, but eases when it falls. This distinction is particularly relevant for the government-spending shock discussed in Section 5, as well as for foreign-interest-rate shocks, where the policy problem may involve cushioning a contraction in credit rather than containing a credit boom. The advantage of capital requirements in Panel A should therefore be interpreted as a greater ability to stabilize credit fluctuations in either direction, rather than simply a stronger capacity to restrain lending.

nisms behind the two instruments are discussed in Section 3.

Panel B of Table 6 turns to the joint use of the two macroprudential instruments.[26] The ranking in Panel A does not translate mechanically into the preferred policy assignment when both instruments are available. For technology, cost-push, and export-demand shocks, the preferred assignment gives reserve requirements the response to credit and capital requirements the response to output. For government-spending and foreign-interest-rate shocks, the ranking reverses. The relevant question is therefore not only which instrument is more effective at stabilizing credit in isolation, but also how the two instruments should divide stabilization tasks when used jointly. This creates scope for complementarity through specialization, since the instrument that is more effective at controlling credit on its own need not always be the one assigned to credit when the central bank considers a broader policy mix.

The government-spending and foreign-interest-rate shocks are especially useful here. In Panel A, these are the shocks for which capital requirements have the largest advantage over reserve requirements when both are assigned to credit fluctuations. In Panel B, they are also the cases in which the preferred assignment gives capital requirements the response to credit. For technology, cost-push, and export-demand shocks, instead, the gains from assigning capital requirements directly to credit are smaller, and the preferred joint assignment gives reserve requirements the response to credit and capital requirements the response to output.

Taken together, these results help explain the broader findings in Section 4. Capital requirements are the stronger tool when the two instruments are used as alternatives to stabilize credit fluctuations. But once both instruments are available, the preferred division of tasks depends on the shock. Reserve requirements are often assigned to credit and capital requirements to output, although this ranking reverses for government-spending and foreign-interest-rate shocks. In this sense, the two instruments are imperfect substitutes when they are asked to do the same task, but they can also complement each other when they are used for different stabilization objectives. The unconditional policy rules should therefore be read as average recommendations across shocks, while the shock-specific exercise shows the heterogeneity behind those averages.

Finally, the robustness exercise in Appendix G shows that the main message does not depend on using the credit-to-output ratio as our financial-stability indicator. When the policy rules and the loss function are instead written in terms of loan-level deviations, the gains from macroprudential policy are larger, and capital requirements remain the stronger tool for leaning against credit fluctuations. What changes is the preferred division of tasks. Stabilizing the level of loans treats all movements in credit as costly, including those that simply reflect changes in economic activity, and this affects how the two instruments are best assigned.

[26] This exercise revisits the strict assignments considered in Policy Block C of Table 5.

7 Conclusions

This paper studies the interaction and effectiveness of two macroprudential instruments under different anchor variables and central bank objectives. We use a small open-economy model with nominal rigidities, financial frictions, and a banking sector subject to reserve requirements and countercyclical capital requirements. The main question is how these instruments should be used over the cycle, and whether they should be viewed as substitutes or complements.

When financial stability is part of the central bank's objective, the gains from including macroprudential tools in the policy mix are significant. These instruments can stabilize both credit and output, and become particularly relevant when macroeconomic and financial conditions call for different policy responses. After a government-spending shock, for example, output and inflation rise, calling for tighter monetary policy, while investment and credit contract, calling for an easing of macroprudential policy.

Importantly, the two instruments work through different channels. Reserve requirements operate mainly through banks' funding conditions, affecting deposit rates and the cost of intermediation, while capital requirements act more directly on lending capacity and loan spreads. These differences shape the response of credit, investment, output, and inflation.

The two instruments can be viewed as imperfect substitutes, while their different transmission channels also create scope for complementarity. When assigned to the same stabilization objective, capital requirements are generally more effective at mitigating credit volatility. This ranking, however, does not determine their preferred assignment under a strict separation of tasks, with one instrument responding to credit and the other to output. In this case, the model suggests assigning reserve requirements to credit and capital requirements to output. This division of tasks is not universal, however, and may change with the nature of the shock hitting the economy.

A natural next step is to introduce an explicit crisis state, so that the role of these instruments can also be studied through crisis prevention and the build-up and release of regulatory buffers.

References

Agénor, P.-R., Alper, K. and da Silva, L. P. (2018), 'External shocks, financial volatility and reserve requirements in an open economy', *Journal of International Money and Finance* **83**, 23–43. 3

Agur, I. (2019), 'Monetary and macroprudential policy coordination among multiple equilibria', *Journal of International Money and Finance* **96**, 192–209. 4

Allen, L. and Rai, A. (1996), 'Operational efficiency in banking: An international comparison', *Journal of Banking & Finance* **20**(4), 655–672. 16

Altavilla, C., Laeven, L. and Peydró, J.-L. (2020), Monetary and macroprudential policy complementarities: evidence from european credit registers, ECB Working Paper Series 2504, European Central Bank. 3

Angelini, P., Neri, S. and Panetta, F. (2014), 'The interaction between capital requirements and monetary policy', *Journal of Money, Credit and Banking* **46**(6), 1073–1112. 3, 15

Angeloni, I., Claessens, S., Seru, A., Steffen, S. and Weder di Mauro, B. (2024), *Geneva 27: Much Money, Little Capital, and Few Reforms: The 2023 banking turmoil*, CEPR Press, Paris and London. 1

Barth, J. R., Lin, C., Ma, Y., Seade, J. and Song, F. M. (2013), 'Do bank regulation, supervision and monitoring enhance or impede bank efficiency?', *Journal of Banking & Finance* **37**(8), 2879–2892. 16

Bénassy-Quéré, A. (2024), 'Banks' liquidity in volatile macroeconomic and market environments'. Speech at the Joint Conference by the Research Group of the Basel Committee on Banking Supervision, the Bank for International Settlements and the Centre for Economic Policy Research, Basel.
URL: *https://www.bis.org/speeches/20240523-banks-liquidity-volatile-macroeconomic-and-market-environments* 1

Beneš, J. and Kumhof, M. (2015), 'Risky bank lending and countercyclical capital buffers', *Journal of Economic Dynamics and Control* **58**, 58–80. 3

Bernanke, B. S., Gertler, M. and Gilchrist, S. (1999), The financial accelerator in a quantitative business cycle framework, *in* J. B. Taylor and M. Woodford, eds, 'Handbook of Macroeconomics', Vol. 1, Elsevier, pp. 1341–1393. 6, 10

Borio, C. E. V. and Drehmann, M. (2009), 'Assessing the risk of banking crises–revisited', *BIS Quarterly Review* pp. 29–46. 16

Borio, C. and Lowe, P. (2002), 'Assessing the risk of banking crises', *BIS Quarterly Review* pp. 43–54. 16

Bustamante, C. and Hamann, F. (2015), 'Countercyclical reserve requirements in a

heterogeneous-agent and incomplete financial markets economy', *Journal of Macroeconomics* **46**, 55–70. 3

Cantú, C., Gondo, R. and Martínez, B. (2024), Reserve requirements as a financial stability instrument, BIS Working Papers 1182, Bank for International Settlements. 1, 3

Carvalho, F. A. and Castro, M. R. (2017), 'Macroprudential policy transmission and interaction with fiscal and monetary policy in an emerging economy: a DSGE model for Brazil', *Macroeconomics and Finance in Emerging Market Economies* **10**(3), 215–259. 4

Cecchetti, S. G. and Kohler, M. (2014), 'When capital adequacy and interest rate policy are substitutes (and when they are not)', *International Journal of Central Banking* **10**(3), 205–231. 3

Cerutti, E., Claessens, S. and Laeven, L. (2017), 'The use and effectiveness of macroprudential policies: New evidence', *Journal of Financial Stability* **28**, 203–224. 3

Christensen, I. and Dib, A. (2008), 'The financial accelerator in an estimated New Keynesian model', *Review of Economic Dynamics* **11**(1), 155–178. 16

Christoffel, K. P., Coenen, G. and Warne, A. (2008), The New Area-Wide Model of the euro area: a micro-founded open-economy model for forecasting and policy analysis, ECB Working Paper Series 944, European Central Bank. 16, 17

Clancy, D. and Merola, R. (2017), 'Countercyclical capital rules for small open economies', *Journal of Macroeconomics* **54**, 332–351. 3

Cordella, T., Federico, P., Végh, C. and Vuletin, G. (2014), *Reserve requirements in the brave new macroprudential world*, World Bank, Washington, DC. 1, 3

Darracq Pariès, M., Karadi, P., Kok, C. and Nikolov, K. (2022), 'The impact of capital requirements on the macroeconomy: lessons from four macroeconomic models of the euro area', *International Journal of Central Banking* **18**(5), 197–246. 3

Darracq Pariès, M., Sørensen, C. K. and Rodriguez-Palenzuela, D. (2011), 'Macroeconomic propagation under different regulatory regimes: Evidence from an estimated DSGE model for the Euro area', *International Journal of Central Banking* **7**(4), 49–113. 3

de Cos, P. H. (2023), The role of macroprudential policy in the stabilisation of macrofinancial fluctuations. conference on financial stability/banco de portugal, lisbon (portugal), 2 october 2023.(el papel de la política macroprudencial en la estabilización de las fluctuaciones macrofinancieras. conferencia de estabilidad financiera/banco de portugal, lisboa (portugal), 2 de octubre de 2023), *in* 'Conference on Financial Stability/Banco de Portugal, Lisbon (Portugal)', Vol. 2. 1

De Paoli, B. and Paustian, M. (2017), 'Coordinating monetary and macroprudential policies', *Journal of Money, Credit and Banking* **49**(2-3), 319–349. 3

Drehmann, M., Borio, C. E. V., Gambacorta, L., Jiménez, G. and Trucharte, C. (2010), Countercyclical capital buffers: exploring options, BIS Working Papers 317, Bank for International Settlements. 16

Drehmann, M., Borio, C. E. V. and Tsatsaronis, K. (2011), 'Anchoring countercyclical capital buffers: the role of credit aggregates', *International Journal of Central Banking* **7**(4), 189–240. 3

Drehmann, M. and Gambacorta, L. (2012), 'The effects of countercyclical capital buffers on bank lending', *Applied Economics Letters* **19**(7), 603–608. 3

Ferreira, L. N. and Nakane, M. I. (2018), 'Macroprudential policy in a DSGE model: anchoring the countercyclical capital buffer', *Economics Bulletin* **38**(4), 2345–2352. 3

Ferrero, A., Harrison, R. and Nelson, B. (2024), 'House price dynamics, optimal LTV limits and the liquidity trap', *The Review of Economic Studies* **91**(2), 940–971. 15

Frache, S., García-Cicco, J. and Ponce, J. (2023), 'Countercyclical prudential tools in an estimated DSGE model', *Latin American Journal of Central Banking* **4**(3), 100095. 3

García Revelo, J. D. and Levieuge, G. (2022), 'When could macroprudential and monetary policies be in conflict?', *Journal of Banking & Finance* **139**, 106484. 3

Gelain, P. and Ilbas, P. (2017), 'Monetary and macroprudential policies in an estimated model with financial intermediation', *Journal of Economic Dynamics and Control* **78**, 164–189. 3

Gerali, A., Neri, S., Sessa, L. and Signoretti, F. M. (2010), 'Credit and banking in a DSGE model of the Euro area', *Journal of Money, Credit and Banking* **42**(s1), 107–141. 4, 6, 16

Gertler, M., Gilchrist, S. and Natalucci, F. M. (2007), 'External constraints on monetary policy and the financial accelerator', *Journal of Money, Credit and Banking* **39**(2-3), 295–330. 10

Glocker, C. and Towbin, P. (2012), 'Reserve requirements for price and financial stability: When are they effective?', *International Journal of Central Banking* **8**(1), 65–114. 3, 4, 8, 16

Glocker, C. and Towbin, P. (2015), 'Reserve requirements as a macroprudential instrument–empirical evidence from Brazil', *Journal of Macroeconomics* **44**, 158–176. 3

Hecker, D., Jang, H., Rubio, M. and Verona, F. (2024), Robust design of countercyclical capital buffer rules, Research Discussion Papers 9/2024, Bank of Finland. 3, 15

Koch, C., Richardson, G. and Van Horn, P. (2025), 'Countercyclical capital buffers: a cautionary tale', *Journal of Money, Credit and Banking* **57**(4), 793–832. 3

Laeven, L., Maddaloni, A. and Mendicino, C. (2022), Monetary policy, macroprudential policy and financial stability, ECB Working Paper Series 2647, European Central Bank.

3

Lozej, M., Onorante, L. and Rannenberg, A. (2023), 'Countercyclical capital regulation in a small open economy DSGE model', *Macroeconomic Dynamics* **27**(5), 1230–1267. 3

Malherbe, F. (2020), 'Optimal capital requirements over the business and financial cycles', *American Economic Journal: Macroeconomics* **12**(3), 139–174. 3

Mendicino, C., Nikolov, K., Suárez, J. and Supera, D. (2018), 'Optimal dynamic capital requirements', *Journal of Money, Credit and Banking* **50**(6), 1271–1297. 3

Peura, S. and Jokivuolle, E. (2004), 'Simulation based stress tests of banks' regulatory capital adequacy', *Journal of Banking & Finance* **28**(8), 1801–1824. 16

Quint, D. and Rabanal, P. (2014), 'Monetary and macroprudential policy in an estimated DSGE model of the Euro area', *International Journal of Central Banking* **10**(2), 169–236. 3

Rubio, M. and Carrasco-Gallego, J. A. (2014), 'Macroprudential and monetary policies: Implications for financial stability and welfare', *Journal of Banking & Finance* **49**, 326–336. 3

Rubio, M. and Carrasco-Gallego, J. A. (2015), 'Macroprudential and monetary policy rules: a welfare analysis', *The Manchester School* **83**(2), 127–152. 3

Schmitt-Grohé, S. and Uribe, M. (2007), 'Optimal simple and implementable monetary and fiscal rules', *Journal of Monetary Economics* **54**(6), 1702–1725. 3

Van der Ghote, A. (2021), 'Interactions and coordination between monetary and macroprudential policies', *American Economic Journal: Macroeconomics* **13**(1), 1–34. 3

Appendix

A The log-linearized Equations

Unless otherwise stated, hatted real quantities are log deviations from steady state, and hatted gross interest rates are fractional deviations from their steady state gross rates. The reserve-ratio variables $\tilde{\varsigma}_t$, $\tilde{\tau}_t^R$ and the capital requirement variable $\tilde{\tau}_t^K$ are level deviations in ratio units; for example, 0.01 denotes a one percentage point increase in the corresponding requirement.

Households

- Consumption-saving decision:

$$E_t\hat{C}_{t+1} - \hat{C}_t = \hat{i}_t^D - E_t\hat{\pi}_{t+1}$$

- Uncovered interest parity condition:

$$\hat{i}_t^D + \psi_B\hat{B}_t = \hat{i}_t^* + E_t\Delta\hat{s}_{t+1}$$

- Labor supply:

$$\hat{w}_t = \phi\hat{h}_t + \hat{C}_t$$

Deposit Units

- Reserve requirements:

$$\tilde{\varsigma}_t = \tilde{\tau}_t^R$$

- Deposit rate:

$$\hat{i}_t^D = \frac{i^{IB}}{i^D}\hat{i}_t^{IB} - \frac{i^{IB} - i^R}{i^D}\tilde{\tau}_t^R$$

- Reserves:

$$\hat{R}_t = \hat{D}_t + \frac{\tilde{\varsigma}_t}{\varsigma}$$

- Interbank lending:

$$\hat{D}_t^{IB} = \hat{D}_t - \frac{\tilde{\varsigma}_t}{1-\varsigma}$$

Lending Units

- Balance Sheet:

$$\hat{L}_t = \frac{D^{IB}}{L}\hat{D}_t^{IB} + \frac{K^b}{L}\hat{K}_t^b$$

- Bank capital dynamics

$$\hat{K}_t^b = (1-\delta^b)\hat{K}_{t-1}^b + \delta^b\hat{\Pi}_t$$

- Effective interest rate

$$\hat{i}_t^E = \frac{i^{IB}}{i^E}\hat{i}_t^{IB} - \kappa\frac{(K^b/L)^2}{i^E}\left[(3(K^b/L)-2\tau^K)(\hat{K}_t^b - \hat{L}_t) - \tilde{\tau}_t^K\right]$$

- Profits

$$\begin{aligned}\delta^b\hat{\Pi}_t = &\frac{i^L}{K^b/L}\left(\hat{i}_{t-1}^L - \hat{\pi}_t\right) + \frac{i^L-1}{K^b/L}\hat{L}_{t-1}\\ &-\frac{1-K^b/L}{K^b/L}\left[i^{IB}\left(\hat{i}_{t-1}^{IB} - \hat{\pi}_t\right) + (i^{IB}-1)\hat{D}_{t-1}^{IB}\right]\\ &-\frac{\kappa}{2}(K^b/L-\tau^K)^2\hat{K}_{t-1}^b - \kappa(K^b/L-\tau^K)\left[(K^b/L)(\hat{K}_{t-1}^b - \hat{L}_{t-1}) - \tilde{\tau}_{t-1}^K\right].\end{aligned}$$

Financial contract

- Leverage and external finance premium:

$$E_t\hat{r}_{t+1}^K - \hat{i}_t^E + E_t\hat{\pi}_{t+1} = \eta(\hat{Q}_t + \hat{K}_t - \hat{N}_t)$$

- Loan rate (nominal and real):

$$\hat{r}_t^L = \hat{Q}_t + \hat{K}_t + E_t\hat{r}_{t+1}^K - \hat{L}_t$$

$$\hat{i}_t^L = \hat{r}_t^L + E_t\hat{\pi}_{t+1}$$

Entrepreneurs

- Balance Sheet:

$$\hat{Q}_t + \hat{K}_t = \epsilon_L\hat{L}_t + (1-\epsilon_L)\hat{N}_t$$

- Net Worth:

$$\hat{N}_t = \nu\hat{N}_{t-1} + (1-\nu)(\hat{Q}_{t-1} + \hat{K}_{t-1}) + \hat{r}_t^K + \nu\frac{\epsilon_L}{1-\epsilon_L}(\hat{r}_t^K - (\hat{i}_{t-1}^L - \hat{\pi}_t))$$

Intermediate Goods Producers

- Production function:
$$\hat{y}_t = \hat{\xi}_t^A + \alpha \hat{K}_{t-1} + (1-\alpha)\hat{h}_t$$
- Marginal costs:
$$\hat{mc}_t = \alpha \hat{z}_t + (1-\alpha)\hat{W}_t - \hat{\xi}_t^A$$
- Cost minimization:
$$\hat{h}_t + \hat{W}_t = \hat{z}_t + \hat{K}_{t-1}$$
- Price setting:
$$\hat{\pi}_t^d = \beta E_t \hat{\pi}_{t+1}^d + \frac{(1-\theta)(1-\theta\beta)}{\theta} \hat{mc}_t + \hat{\xi}_t^{CP}$$

Capital Goods Producers

- Investment Demand:
$$\hat{Q}_t = \chi(\hat{I}_t - \hat{K}_{t-1})$$
- Price of capital:
$$\hat{r}_t^K + \hat{Q}_{t-1} = \frac{MPK}{r^K}\hat{z}_t + \frac{1-\delta}{r^K}\hat{Q}_t$$
where MPK is the marginal product of capital.
- Capital dynamics:
$$\hat{K}_t = (1-\delta)\hat{K}_{t-1} + \delta \hat{I}_t$$

Monetary and Macroprudential Policy

- Taylor-rule:
$$\hat{i}_t^{IB} = \phi_{\pi,i}\hat{\pi}_t + \phi_{Y,i}\hat{Y}_t + \phi_{l,i}\hat{l}_t$$
- Reserve requirements:
$$\tilde{\tau}_t^R = \phi_{Y,R}\hat{Y}_t + \phi_{l,R}\hat{l}_t$$
- Capital requirements:
$$\tilde{\tau}_t^K = \phi_{Y,K}\hat{Y}_t + \phi_{l,K}\hat{l}_t$$

Market Clearing

- Goods market:

$$\hat{Y}_t = \gamma(c_y\hat{C}_t + i_y\hat{I}_t + g_y\hat{G}_t + (1-\gamma)\hat{\epsilon}_t) + (1-\gamma)(\hat{\epsilon}_t + \hat{X}_t)$$

- Balance of payments

$$\hat{B}_t = \hat{Y}_t - (c_y \hat{C}_t + i_y \hat{I}_t + g_y \hat{G}_t + (1-\gamma)\hat{\epsilon}_t) + i^* \hat{B}_{t-1}$$

- Real exchange rate:

$$\hat{\epsilon}_t - \hat{\epsilon}_{t-1} = \Delta \hat{s}_t - \hat{\pi}_t^d$$

- CPI inflation rate:

$$\hat{\pi}_t = \gamma \hat{\pi}_t^d + (1-\gamma)\Delta \hat{s}_t$$

B Calibration

The calibration of the remaining model parameters is reported in Tables 1 and 2 in Section 2.6.

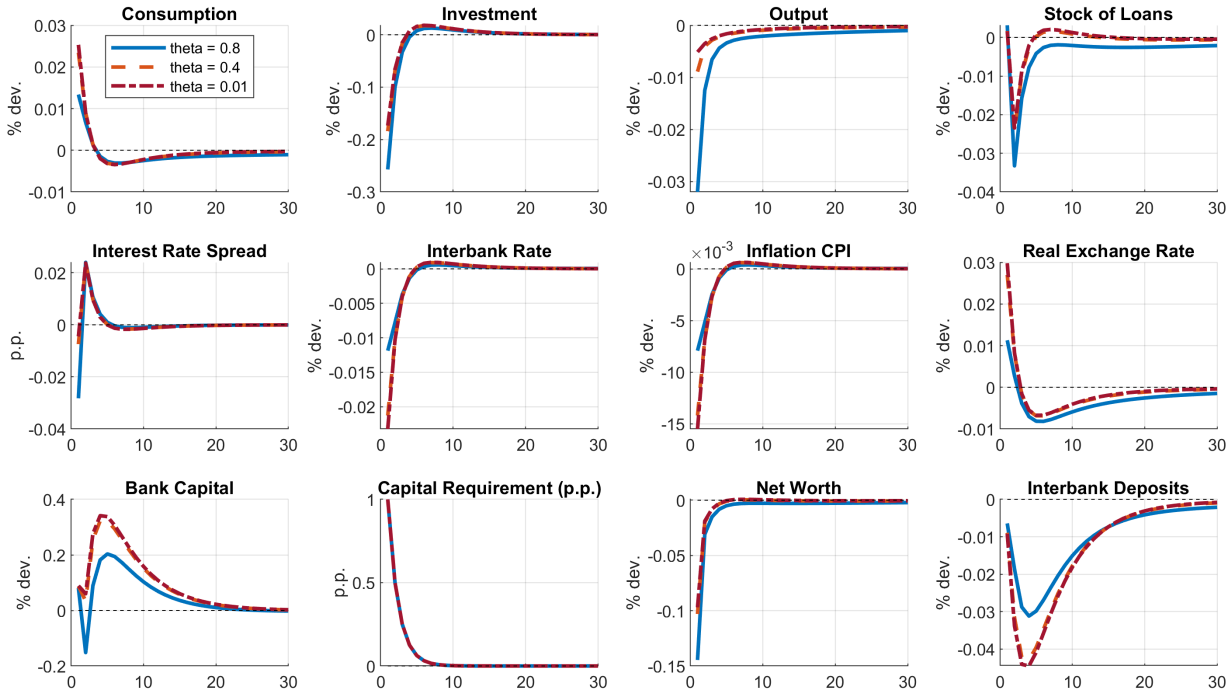
Table 7: Steady-State Values

Param.	Value	Description
MPK	0.047	Marginal Product of Capital
r^K	1.022	Gross Real Return on Capital
i^R	1.0006	Gross Interest Rate on Reserves
i^D	1.0152	Gross Deposit Rate
i^{IB}	1.0156	Gross Interbank Rate
i^*	1.0025	Gross Foreign Interest Rate
c_y	0.55	Consumption-Output Ratio
i_y	0.22	Investment-Output Ratio
g_y	0.23	Government Spending-Output Ratio
ς	0.10	Effective Reserve Ratio
$\bar{\tau}^R$	0.09	Required Reserve Ratio
K^b/L	0.11	Bank Capital-to-Loans Ratio
$\bar{\tau}^K$	0.09	Target Capital Requirement Ratio
i^E	1.0147	Gross Effective Lending Rate
i^L	1.0255	Gross Contractual Loan Rate

Note: Gross rates are quarterly gross rates. Ratios are reported in levels. The effective lending rate i^E and contractual loan rate i^L are implied by the banking block using the structural parameters in Table 2.

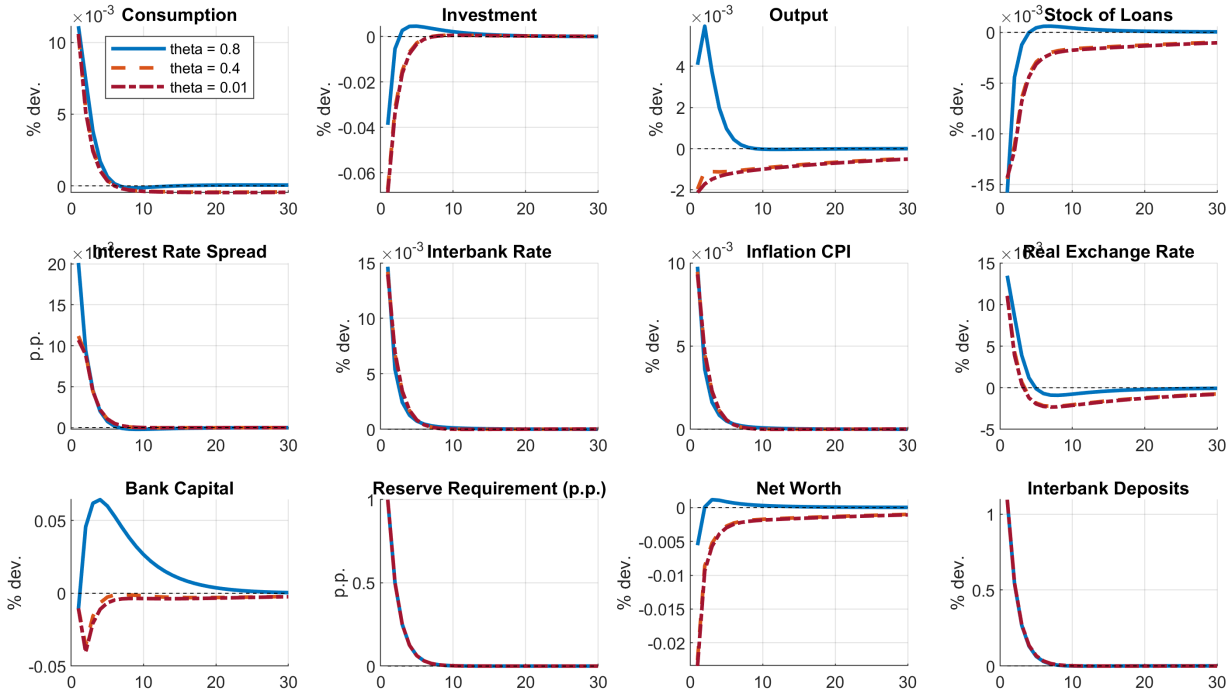
C Impulse Response Functions

Figure 7: Capital Requirement Shock and the degree of Price Stickiness



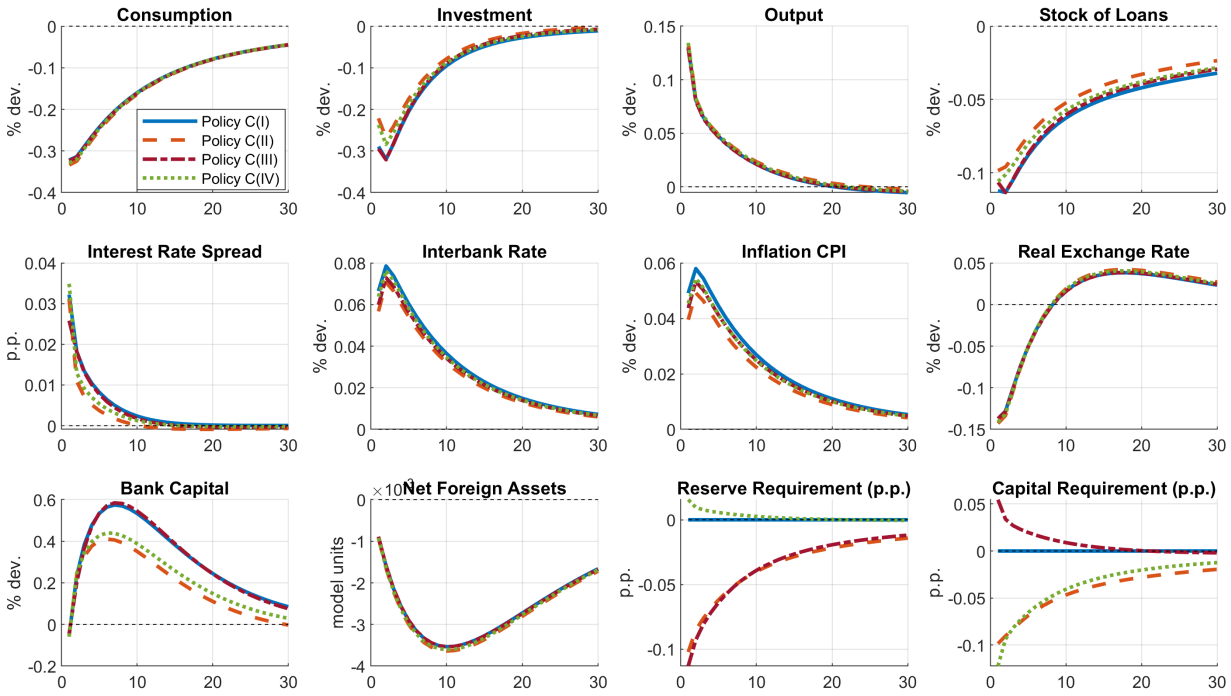
Note: The figure reports quarterly impulse responses to a 1 p.p. increase in capital requirements, considering a scenario with a financial accelerator mechanism and different values of θ. Monetary policy is specified by an interest rate rule for the interbank interest rate as defined in section 2.5 with $\phi_{\pi,i} = 1.5$ and the other coefficients equal 0. The y-axis denotes the deviation from the steady state either in percent or in percentage points.

Figure 8: Reserve Requirement Shock and the degree of Price Stickiness



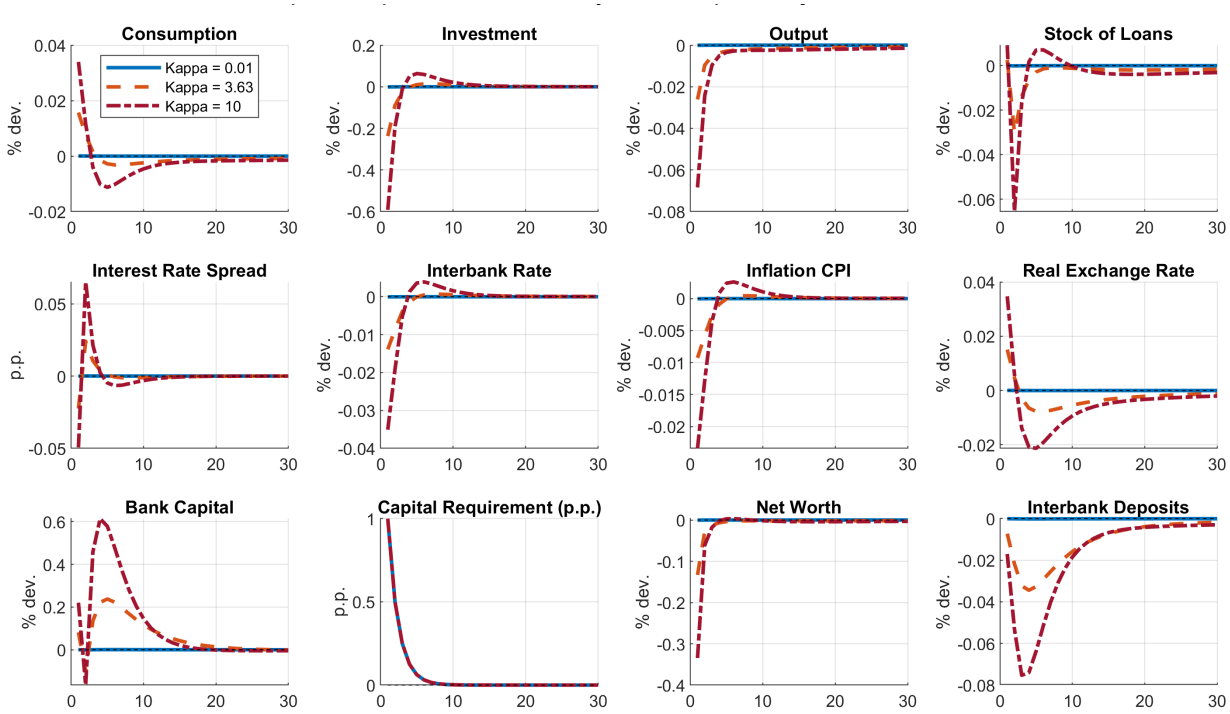
Note: The figure reports quarterly impulse responses to a 1 p.p. increase in reserve requirements, considering a scenario with a financial accelerator mechanism and different values of θ. Monetary policy is specified by an interest rate rule for the interbank interest rate as defined in section 2.5 with $\phi_{\pi,i} = 1.5$ and the other coefficients equal 0. The y-axis denotes the deviation from the steady state either in percent or in percentage points.

Figure 9: Government shock under Policy Block C and a Financial Stability Objective



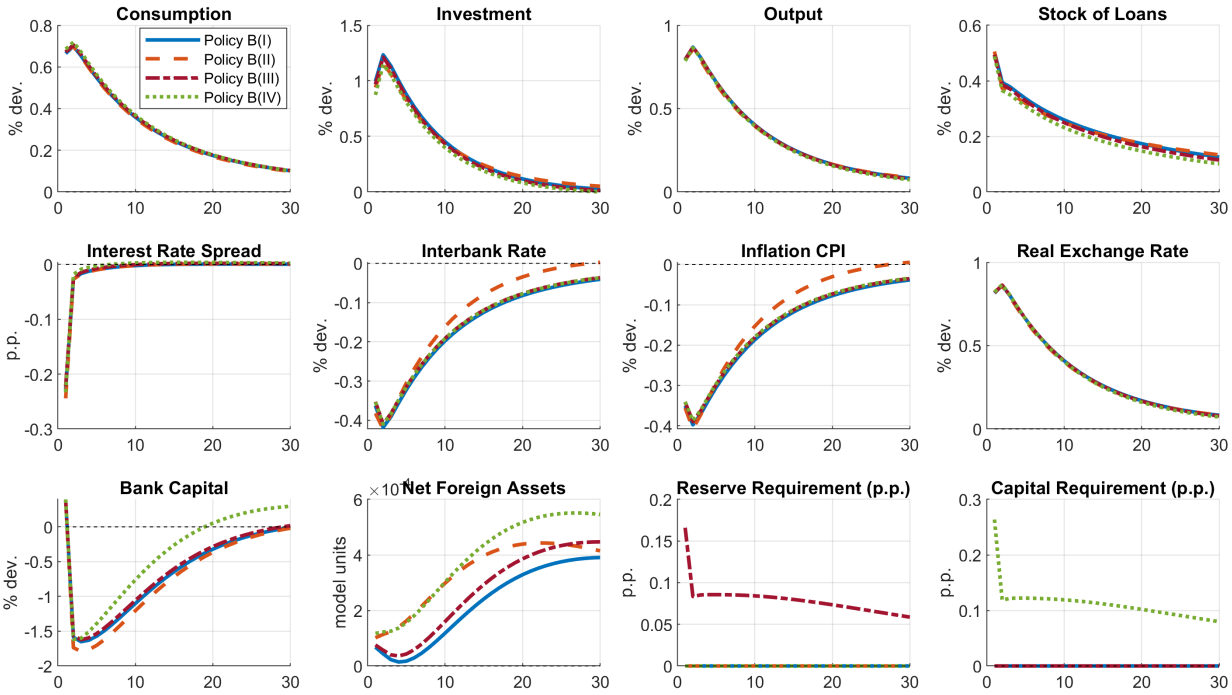
Note: The figure reports quarterly impulse responses to a 1-std deviation increase in government expenditure, considering a scenario with a financial accelerator mechanism and different policy rules, as described in the legend. The y-axis denotes the deviation from the steady state either in percent or in percentage points.

Figure 10: Capital requirement shock under different degrees of bank capital adjustment cost



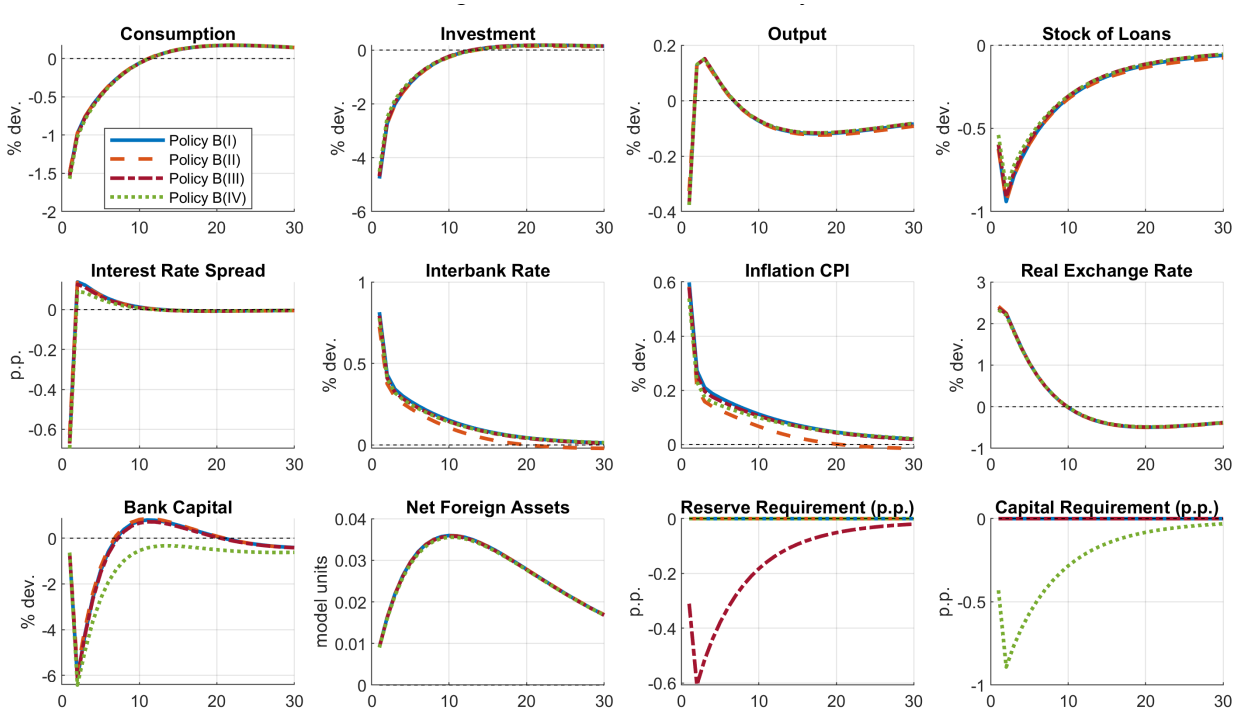
Note: The figure reports quarterly impulse responses to a 1 p.p. increase in capital requirements, considering different values for κ, the adjustment cost on bank capital. Monetary policy is specified by an interest rate rule for the interbank interest rate as defined in section 2.5 with $\phi_{\pi,i} = 1.5$ and the other coefficients equal 0. The y-axis denotes the deviation from the steady state either in percent or in percentage points.

Figure 11: Technology shock under different Policy Rules and a Financial Stability Objective



Note: The figure reports quarterly impulse responses to a positive 1-std deviation technology shock, considering a scenario with a financial accelerator mechanism and different policy rules, as described in the legend. The y-axis denotes the deviation from the steady state either in percent or in percentage points.

Figure 12: Foreign Interest Rate shock under different Policy Rules and a Financial Stability Objective



Note: The figure reports quarterly impulse responses to a positive 1-std deviation foreign interest rate shock, considering a scenario with a financial accelerator mechanism and different policy rules, as described in the legend. The y-axis denotes the deviation from the steady state either in percent or in percentage points.

D Robustness to Instrument-penalty Weights

This appendix examines whether the optimized policy coefficients in Policy B are sensitive to the weight placed on macroprudential-instrument volatility in the augmented OSR objective. The core financial-stability loss remains $\mathcal{L}^{FS}(\Phi_n)$, as in equation (32). For each value of λ_τ, however, the active coefficients are reoptimized using the augmented objective in equation (33). The tables report both the core loss and the augmented objective, so that the stabilization trade-off can be read separately from the instrument-smoothing term.

The exercise is run separately for the two one-instrument macroprudential policies in block B. Policy B(III) allows reserve requirements to respond to output and to the credit-to-output gap, while capital requirements are inactive. Policy B(IV) performs the analogous exercise for capital requirements.

Table 8: Policy B(III): Robustness to Reserve-requirement Penalty Weights

	λ_τ					
	0	0.050	0.100	0.250	0.500	1.000
$\phi_{\pi,i}$	14.89	1.513	1.501	1.493	1.490	1.488
$\phi_{Y,i}$	7.910	0.205	0.202	0.201	0.200	0.200
$\phi_{Y,R}$	29.62	0.844	0.450	0.189	0.096	0.049
$\phi_{l,R}$	30.75	1.231	0.638	0.262	0.132	0.067
$10^4 \times \mathcal{L}^{FS}(\Phi_n)$	9.614	12.90	13.22	13.45	13.53	13.58
$10^4 \times Q_n^{FS}(\Phi_n)$	9.614	13.23	13.41	13.53	13.58	13.60
$\Delta\mathcal{L}^{FS}$ relative to Policy B(I)	-29.43	-5.274	-2.960	-1.287	-0.665	-0.338
$sd(\tilde{\tau}_t^R)$, p.p.	30.38	2.568	1.385	0.585	0.299	0.151

Notes: The table reports the optimized coefficients for Policy B(III) as the macroprudential-instrument penalty weight λ_τ changes. The core loss $\mathcal{L}^{FS}$ excludes the penalty term, while Q_n^{FS} is the augmented OSR objective minimized by the optimizer. The final row reports the standard deviation of the active reserve-requirement instrument in percentage points.

Table 9: Policy B(IV): Robustness to Capital-requirement Penalty Weights

	λ_τ					
	0	0.050	0.100	0.250	0.500	1.000
$\phi_{\pi,i}$	11.41	1.754	1.619	1.541	1.514	1.501
$\phi_{Y,i}$	4.897	0.311	0.252	0.220	0.209	0.204
$\phi_{Y,K}$	30.17	1.347	0.712	0.303	0.156	0.080
$\phi_{l,K}$	33.84	1.996	0.994	0.403	0.204	0.103
$10^4 \times \mathcal{L}^{FS}(\Phi_n)$	8.924	12.06	12.72	13.21	13.41	13.51
$10^4 \times Q_n^{FS}(\Phi_n)$	8.924	12.76	13.14	13.41	13.51	13.57
$\Delta\mathcal{L}^{FS}$ relative to Policy B(I)	-34.49	-11.44	-6.623	-3.009	-1.594	-0.824
$sd(\tilde{\tau}_t^K)$, p.p.	19.28	3.728	2.044	0.887	0.461	0.236

Notes: The table reports the optimized coefficients for Policy B(IV) as the macroprudential-instrument penalty weight λ_τ changes. The core loss $\mathcal{L}^{FS}$ excludes the penalty term, while Q_n^{FS} is the augmented OSR objective minimized by the optimizer. The final row reports the standard deviation of the active capital-requirement instrument in percentage points.

E Stabilization Trade-Off Exercise Details

This appendix describes the construction of the stabilization frontiers reported in Section 4.3. For this exercise we hold constant the model, calibration, shock processes, and optimal-simple-rule (OSR) problem as the baseline policy analysis, but vary the relative weight assigned to credit stabilization along the points of the frontier.

For each policy arrangement n and value of λ_l, we choose the policy coefficients Φ_n to minimize

$$Q_n(\Phi_n; \lambda_l) = \text{var}(\hat{\pi}_t) + \text{var}(\hat{Y}_t) + \lambda_l \,\text{var}(\hat{l}_t) + \lambda_\tau \left[\text{var}(\tilde{\tau}_t^R) + \text{var}(\tilde{\tau}_t^K)\right], \tag{36}$$

where $\lambda_Y = 1$ and $\lambda_\tau = 0.10$. Unconditional variances are computed with all five calibrated macroeconomic shocks active: technology, cost-push, government spending, export demand, and the foreign interest rate shock.

We consider four sets of available policy coefficients:

$$\begin{aligned}
\Phi_{\text{B(I)}} &= \{\phi_{\pi,i}, \phi_{Y,i}\},\\
\Phi_{\text{B(III)}} &= \{\phi_{\pi,i}, \phi_{Y,i}, \phi_{Y,R}, \phi_{l,R}\},\\
\Phi_{\text{B(IV)}} &= \{\phi_{\pi,i}, \phi_{Y,i}, \phi_{Y,K}, \phi_{l,K}\},\\
\Phi_{\text{RR+CR}} &= \{\phi_{\pi,i}, \phi_{Y,i}, \phi_{Y,R}, \phi_{l,R}, \phi_{Y,K}, \phi_{l,K}\}.
\end{aligned}$$

The first three correspond to Policies B(I), B(III), and B(IV) in Table 4. The last specification extends Policies B(III) and B(IV) by allowing both macroprudential instruments to respond simultaneously to output and the credit-to-output gap.

We solve the OSR problem over

$$\lambda_l \in \{0,\ 0.05,\ 0.10,\ 0.25,\ 0.50,\ 1,\ 2,\ 4,\ 4\sqrt{2},\ 8,\ 8\sqrt{2},\ 16,\ 16\sqrt{2},\ 32\}.$$

The active coefficients are re-optimized separately at every grid point, without imposing additional coefficient bounds. For each line, Figure 5 only shows coordinates obtained from solutions that weakly dominate other results within the same policy set. This allows the figure to be interpreted as a frontier, since other solutions might not be included because their coordinates are "interior" (weakly located to the right of the line).

Finally, we report solutions using $\lambda_l \leq 8$ as an upper limit, since the absence of coefficient bounds can generate very large policy responses when the policymaker places extreme weight on credit stabilization.

F Shock-specific OSR Procedure

This appendix describes the shock-specific optimal simple-rule exercise used in Section 6. The exercise is run separately for each shock $u \in \mathcal{U} \equiv \{\xi^A, \xi^{CP}, G, X, i^*\}$.

The corresponding shock processes are the technology shock ξ_t^A, the cost-push shock ξ_t^{CP}, the government spending shock G_t, the foreign export demand shock X_t, and the foreign interest rate shock i_t^*. Policy shocks to reserve requirements and capital requirements are excluded. For each shock u, the OSR problem is solved with only the innovation associated with that shock active.

The shock-specific core loss is given by equation (34). The objective minimized in the numerical optimization augments this core loss with an instrument-volatility penalty:

$$Q_{n,u}^{FS}(\Phi_n) = \mathcal{L}_u^{FS}(\Phi_n) + \lambda_\tau E_u\left[(\tilde{\tau}_t^R)^2 + (\tilde{\tau}_t^K)^2\right], \tag{37}$$

where $\lambda_\tau = 0.10$. The penalty applies only to reserve and capital requirements. If one

of these instruments is inactive under a given policy arrangement, its deviation from steady state is zero by construction. The penalty is used during optimization, but the reported gains rank policy arrangements using the core loss $\mathcal{L}_u^{FS}(\Phi_n)$. For policy rule n and benchmark rule b, the gain is given by equation (35).

Panel A of Table 6 compares three policy arrangements. The benchmark is

$$i_t^{IB} \to (\hat{\pi}_t, \hat{Y}_t),$$

while the two alternatives add either reserve requirements or capital requirements as direct responses to the credit-to-output gap:

$$\begin{aligned} i_t^{IB} &\to (\hat{\pi}_t, \hat{Y}_t), \qquad \tau_t^R \to \hat{l}_t, \\ i_t^{IB} &\to (\hat{\pi}_t, \hat{Y}_t), \qquad \tau_t^K \to \hat{l}_t. \end{aligned}$$

Panel B compares strict assignments. The benchmark is

$$i_t^{IB} \to \hat{\pi}_t,$$

and the two alternatives assign one macroprudential instrument to output and the other to the credit-to-output gap:

$$\begin{aligned} i_t^{IB} &\to \hat{\pi}_t, \qquad \tau_t^R \to \hat{l}_t, \qquad \tau_t^K \to \hat{Y}_t, \\ i_t^{IB} &\to \hat{\pi}_t, \qquad \tau_t^R \to \hat{Y}_t, \qquad \tau_t^K \to \hat{l}_t. \end{aligned}$$

As an additional diagnostic, we also considered a broader macroprudential rule in which both reserve requirements and capital requirements respond to output and to the credit-to-output gap. This broader rule delivers larger gains than the strict assignments, as expected, because it gives both macroprudential instruments additional feedback margins. We therefore treat this rule as an upper-bound diagnostic.

G Loan-level Alternative OSR Specification

This appendix reports a robustness exercise in which the credit term in the policy rules and in the financial-stability loss is the loan-level deviation, $\hat{L}_t$, rather than the credit-to-output gap. The aim is to assess whether the main OSR results depend on normalizing credit by output.

The policy rules in this appendix are written as

$$\hat{i}_t^{IB} = \phi_{\pi,i}\hat{\pi}_t + \phi_{Y,i}\hat{Y}_t + \phi_{L,i}\hat{L}_t, \tag{38}$$

$$\tilde{\tau}_t^R = \phi_{Y,R}\hat{Y}_t + \phi_{L,R}\hat{L}_t, \tag{39}$$

$$\tilde{\tau}_t^K = \phi_{Y,K}\hat{Y}_t + \phi_{L,K}\hat{L}_t. \tag{40}$$

Only the coefficients active in a given policy arrangement are optimized; all inactive coefficients are set to zero. The price-stability loss is unchanged:

$$\mathcal{L}^{PS}(\Phi_n) = \mathrm{var}(\hat{\pi}_t) + \mathrm{var}(\hat{Y}_t). \tag{41}$$

The alternative financial-stability loss replaces the credit-to-output gap with loan-level deviations:

$$\mathcal{L}^{FS,L}(\Phi_n) = \mathrm{var}(\hat{\pi}_t) + \mathrm{var}(\hat{Y}_t) + \mathrm{var}(\hat{L}_t). \tag{42}$$

When reserve or capital requirements are active, the numerical OSR objective augments the relevant core loss with the same macroprudential-instrument volatility penalty used in the main tables:

$$Q_n^{j,L}(\Phi_n) = \mathcal{L}^{j,L}(\Phi_n) + \lambda_\tau E\left[(\tilde{\tau}_t^R)^2 + (\tilde{\tau}_t^K)^2\right], \qquad \lambda_\tau = 0.10, \tag{43}$$

where $j \in \{PS, FS\}$. The tables report the core loss, excluding the penalty term, so that the entries are comparable within each block.

Table 10: Alternative Loan-level Specification: Optimal Policy Rules under a Price-stability Objective

	Policy A(I)	Policy A(II)	Policy A(III)
Interest-rate rule	$\hat{\pi}_t, \hat{Y}_t$	$\hat{\pi}_t, \hat{Y}_t$	$\hat{\pi}_t, \hat{Y}_t$
Macroprudential rule	—	$\tau_t^R \to \hat{L}_t$	$\tau_t^K \to \hat{L}_t$
$\phi_{\pi,i}$	2.532	2.511	2.615
$\phi_{Y,i}$	0.653	0.638	0.669
$\phi_{L,R}$	—	0.251	—
$\phi_{L,K}$	—	—	0.391
$10^4 \times \mathcal{L}^{PS}(\Phi_n)$	7.6156	7.5134	7.3782
$\Delta\mathcal{L}^{PS}$ relative to benchmark	—	−1.34%	−3.12%

Notes: This table reruns policy block A with macroprudential rules responding to loan-level deviations $\hat{L}_t$ rather than the credit-to-output gap. The loss remains the core price-stability loss, $\text{var}(\pi_t) + \text{var}(Y_t)$. A dash denotes that the coefficient is not active.

Table 11: Alternative Loan-level Specification: Optimal Policy Rules under a Financial-stability Objective, Policy Block B

	Policy B(I)	Policy B(II)	Policy B(III)	Policy B(IV)
Interest-rate rule	$\hat{\pi}_t, \hat{Y}_t$	$\hat{\pi}_t, \hat{Y}_t, \hat{L}_t$	$\hat{\pi}_t, \hat{Y}_t$	$\hat{\pi}_t, \hat{Y}_t$
RR rule	—	—	$\hat{Y}_t, \hat{L}_t$	—
CR rule	—	—	—	$\hat{Y}_t, \hat{L}_t$
$\phi_{\pi,i}$	1.234	1.116	1.254	1.341
$\phi_{Y,i}$	0.161	0.113	0.161	0.170
$\phi_{L,i}$	—	-0.087	—	—
$\phi_{Y,R}$	—	—	0.383	—
$\phi_{L,R}$	—	—	0.818	—
$\phi_{Y,K}$	—	—	—	0.610
$\phi_{L,K}$	—	—	—	1.237
$10^4 \times \mathcal{L}^{FS,L}(\Phi_n)$	13.51	13.26	12.49	11.26
$\Delta\mathcal{L}^{FS,L}$ relative to benchmark	—	−1.85%	−7.54%	−16.65%

Notes: This table reruns policy block B with loan-level credit responses and with $\mathcal{L}^{FS,L} = \text{var}(\pi_t) + \text{var}(Y_t) + \text{var}(L_t)$. A dash denotes that the coefficient is not active.

Table 12: Alternative Loan-level Specification: Optimal Policy Rules under a Financial-stability Objective, Policy Block C

	Policy C(I)	Policy C(II)	Policy C(III)	Policy C(IV)
Interest-rate rule	$\hat{\pi}_t$	$\hat{\pi}_t$	$\hat{\pi}_t$	$\hat{\pi}_t$
RR rule	—	$\hat{Y}_t, \hat{L}_t$	$\hat{L}_t$	$\hat{Y}_t$
CR rule	—	$\hat{Y}_t, \hat{L}_t$	$\hat{Y}_t$	$\hat{L}_t$
$\phi_{\pi,i}$	1.182	1.295	1.234	1.317
$\phi_{Y,R}$	—	0.128	—	0.455
$\phi_{L,R}$	—	0.913	1.015	—
$\phi_{Y,K}$	—	0.734	1.316	—
$\phi_{L,K}$	—	1.339	—	1.889
$10^4 \times \mathcal{L}^{FS,L}(\Phi_n)$	16.21	11.41	12.24	12.11
$\Delta\mathcal{L}^{FS,L}$ relative to benchmark	—	−29.62%	−24.49%	−25.29%

Notes: This table reruns policy block C with loan-level credit responses and with $\mathcal{L}^{FS,L} = \text{var}(\pi_t) + \text{var}(Y_t) + \text{var}(L_t)$. A dash denotes that the coefficient is not active.

Table 13: Alternative Loan-level Specification: Shock-specific Instrument-assignment Gains

Panel A: Which tool is better at leaning against loan-level fluctuations?

Shock	RR $\rightarrow \hat{L}_t$	CR $\rightarrow \hat{L}_t$	Preferred tool
Technology/productivity, ξ_t^A	7.56	16.27	CR
Cost-push, ξ_t^{CP}	1.37	3.70	CR
Government spending, G_t	15.88	29.63	CR
Export demand, X_t	6.62	12.04	CR
Foreign interest rate, i_t^*	6.30	14.69	CR

Panel B: When macroprudential tools split tasks, which assignment is preferred?

Shock	RR $\rightarrow \hat{L}_t$, CR $\rightarrow \hat{Y}_t$	RR $\rightarrow \hat{Y}_t$, CR $\rightarrow \hat{L}_t$	Preferred assignment
Technology/productivity, ξ_t^A	22.61	19.64	RR $\rightarrow \hat{L}_t$, CR $\rightarrow \hat{Y}_t$
Cost-push, ξ_t^{CP}	15.75	33.01	RR $\rightarrow \hat{Y}_t$, CR $\rightarrow \hat{L}_t$
Government spending, G_t	18.56	32.02	RR $\rightarrow \hat{Y}_t$, CR $\rightarrow \hat{L}_t$
Export demand, X_t	9.17	11.94	RR $\rightarrow \hat{Y}_t$, CR $\rightarrow \hat{L}_t$
Foreign interest rate, i_t^*	9.15	15.05	RR $\rightarrow \hat{Y}_t$, CR $\rightarrow \hat{L}_t$

Notes: Entries are percentage reductions in the shock-specific core financial-stability loss $\mathcal{L}_u^{FS,L}$ relative to the benchmark in the same panel and shock. The core loss is $\text{var}(\pi_t) + \text{var}(Y_t) + \text{var}(L_t)$. When reserve or capital requirements are active, the OSR objective includes a macroprudential-instrument volatility penalty with weight $\lambda_\tau = 0.10$, but assignments are ranked using the core loss. RR denotes reserve requirements, CR denotes capital requirements.

The loan-level specification strengthens the quantitative case for macroprudential policy. Under a price-stability objective, reserve and capital requirements now deliver visible gains, whereas the credit-to-output specification delivers only negligible gains. Under financial stability, the loss reductions are also larger, especially in Policy Block C. The ranking of instruments is broadly consistent with the baseline in one important respect: capital requirements remain the stronger direct instrument for leaning against credit fluctuations. The strict assignment results also become more favorable to assigning capital requirements to loan stabilization in most shock-specific cases.

At the same time, the exercise changes the interpretation of the financial target. Stabilizing $\hat{L}_t$ treats all loan-level fluctuations as costly, including movements associated with normal changes in aggregate activity, whereas the baseline credit-to-output specification focuses more directly on credit imbalances relative to the size of the economy.